STAT 110

Introduction to Probability

Personal Notes

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Chapter 2: Conditional Probability

2.8 Pitfalls and Paradoxes

Conditional probability is how we update beliefs when new evidence shows up. The math is simple, but our gut is bad at it. This section is about three ways conditional reasoning goes wrong:

- the **prosecutor's fallacy**: flipping $P(A \mid B)$ into $P(B \mid A)$,
- the **defense attorney's fallacy**: leaving out part of the evidence,
- **Simpson's paradox**: a trend that holds in every subgroup reverses once the groups are pooled.

Every example here comes down to the same few tools, so here they are in one place.

Toolkit for this section

Let $A_1, \ldots, A_n$ be a partition of S, and let E be any event with $P(E) > 0$.

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)} \qquad \text{Bayes' rule}$$

$$\frac{P(A \mid B)}{P(A^c \mid B)} = \frac{P(B \mid A)}{P(B \mid A^c)} \cdot \frac{P(A)}{P(A^c)} \qquad \text{odds form of Bayes}$$

$$P(B) = \sum_{i=1}^{n} P(B \mid A_i)\,P(A_i) \qquad \text{LOTP}$$

$$P(A \mid B, E) = \frac{P(B \mid A, E)\,P(A \mid E)}{P(B \mid E)} \qquad \text{Bayes, extra conditioning}$$

$$P(B \mid E) = \sum_{i=1}^{n} P(B \mid A_i, E)\,P(A_i \mid E) \qquad \text{LOTP, extra conditioning}$$

In words, the odds form says: posterior odds = likelihood ratio $\times$ prior odds. The extra conditioning versions are just the normal versions with E added to the right of every bar.

Prosecutor's Fallacy

Definition 2.1 (Prosecutor's Fallacy)

The **prosecutor's fallacy** is treating $P(A \mid B)$ and $P(B \mid A)$ as if they were the same number. In a trial this usually looks like

$$\underbrace{P(\text{evidence} \mid \text{innocent})}_{\text{what gets reported}} \quad \text{being read as} \quad \underbrace{P(\text{innocent} \mid \text{evidence})}_{\text{what we actually want}}.$$

Intuition. $P(\text{four legs} \mid \text{dog}) \approx 1$, but $P(\text{dog} \mid \text{four legs})$ is nowhere close to 1, since cats, horses and tables also have four legs. The two conditionals answer different questions. Bayes' rule is the bridge between them, and the bridge always goes through the prior $P(A)$. If the prior is tiny, $P(A \mid B)$ can stay small even when $P(B \mid A)$ is huge.

Example 2.2 (A DNA match that is not 99.999% damning)

A crime happens in a city of 500,001 adults, exactly one of whom is guilty. A suspect is found by searching a DNA database, and there is no other evidence against him, so before the test every adult is equally likely to be the culprit. The guilty person matches for sure, and an innocent person matches by chance with probability 1/100,000. The prosecutor says: "an innocent person matches only 1 in 100,000 times, so the suspect is innocent with probability 1/100,000." Find the actual probability that he is guilty.

Solution. Let G be the event that the suspect is guilty and D the event that his DNA matches. Then

$$P(G) = \frac{1}{500{,}001}, \qquad P(D \mid G) = 1, \qquad P(D \mid G^c) = \frac{1}{100{,}000}.$$

The prosecutor reported $P(D \mid G^c)$ and called it $P(G^c \mid D)$. That's the fallacy. Using the odds form of Bayes instead:

$$\begin{aligned} \frac{P(G \mid D)}{P(G^c \mid D)} &= \frac{P(D \mid G)}{P(D \mid G^c)} \cdot \frac{P(G)}{P(G^c)} && \text{odds form of Bayes} \\ &= 100{,}000 \cdot \frac{1}{500{,}000} && \text{plug in} \\ &= \frac{1}{5}. \end{aligned}$$

Converting odds back to a probability,

$$P(G \mid D) = \frac{1/5}{1 + 1/5} = \frac{1}{6} \approx 0.167.$$

So the suspect is *more likely innocent than guilty*, even though the match sounds overwhelming.

Sanity check with counts. Among the 500,000 innocent adults we expect $500{,}000 \cdot \frac{1}{100{,}000} = 5$ chance matches, plus the 1 guilty person. A random person from those 6 matches is guilty with probability 1/6.

Example 2.3 (The Sally Clark case)

In 1998 in the UK, Sally Clark was tried for murder after two of her baby sons died suddenly, each shortly after birth. An expert witness for the prosecution said that the chance of a newborn dying of sudden infant death syndrome (SIDS) is about 1/8500, so the chance of two SIDS deaths in one family is

$$\left(\frac{1}{8500}\right)^2 = \frac{1}{72{,}250{,}000} \approx \frac{1}{73 \text{ million}},$$

and concluded that the probability Clark was innocent was 1 in 73 million. Find the two mistakes in this argument.

Solution. Mistake 1: assuming independence. Squaring 1/8500 is only valid if the two deaths are independent. They almost certainly aren't: genetics and the home environment are shared, so one SIDS death raises the risk for the next baby. To see how much this matters, suppose (only as an illustration) that after one SIDS death the next baby's risk rises to 1/100. Then

$$P(\text{two SIDS deaths}) = P(\text{1st SIDS})\, P(\text{2nd SIDS} \mid \text{1st SIDS}) \qquad \text{multiplication rule}$$

$$= \frac{1}{8500} \cdot \frac{1}{100} = \frac{1}{850{,}000},$$

which is 85 times more likely than the expert's number.

Mistake 2: the prosecutor's fallacy. Let E be the evidence (two infant deaths), I innocence and G guilt. The expert's number, however it's computed, is $P(E \mid I)$. The court needs $P(I \mid E)$. By Bayes' rule and LOTP,

$$P(I \mid E) = \frac{P(E \mid I)\, P(I)}{P(E \mid I)\, P(I) + P(E \mid G)\, P(G)} \qquad \text{Bayes + LOTP}$$

$$= \frac{P(E \cap I)}{P(E \cap I) + P(E \cap G)}. \qquad \text{def. of cond. prob.}$$

Here $P(E \cap I)$ is the chance a family has two SIDS deaths, and $P(E \cap G)$ is the chance of a double murder of infants by a parent. *Both* are tiny. The answer depends on how they compare, not on how small either one is by itself. Write $s = P(E \cap I)$ and $m = P(E \cap G)$. Then

$$P(I \mid E) = \frac{s}{s+m} = \frac{1}{1 + m/s}.$$

Double murder vs. double SIDS, m/s	$\frac{1}{10}$	$\frac{1}{2}$	1
$P(I \mid E)$	$\frac{10}{11} \approx 0.91$	$\frac{2}{3} \approx 0.67$	$\frac{1}{2} = 0.50$

Even if a double murder were exactly as likely as a double SIDS death, the probability of innocence would be 1/2, not 1 in 73 million. The expert's number is only one piece of Bayes' rule; the prior is the other. Clark was convicted, and the conviction was overturned on appeal after she had spent more than three years in prison.

Pitfall. A small $P(\text{evidence} \mid \text{innocent})$ only means the evidence is *surprising* under innocence. If guilt is also surprising, the evidence doesn't separate the two. Always ask: compared to what?

Defense Attorney's Fallacy

Definition 2.4 (Defense Attorney's Fallacy)

The **defense attorney's fallacy** is failing to condition on all the evidence. If we know both B and C happened, the relevant probability is $P(A \mid B, C)$, and reporting $P(A \mid B)$ instead throws away the information in C.

Intuition. Every conditional probability is an average over some population. $P(A \mid B)$ averages over *everyone* in B, and $P(A \mid B, C)$ averages over only the people in both B and C. If C is a rare event that is strongly tied to A, those two populations look nothing alike, so the two probabilities can be wildly different.

Example 2.5 (History of abuse in a murder trial)

A woman is murdered and her husband is put on trial. The prosecution brings up that he has a history of abusing her. The defense says this is irrelevant, because only 1 in 10,000 abusive husbands go on to murder their wives. Take that figure as correct, and assume the following about the relevant population:

- 1 in 10 husbands abuse their wives,
- 1 in 5 murdered wives were killed by their husbands,
- 50% of husbands who murder their wives abused them beforehand,
- if a murdered woman's husband is innocent, the chance he abused her is just the baseline rate.

Is the abuse history really irrelevant?

Solution. Let A be the event that the husband abused his wife, G that he is guilty, and M that the wife was murdered. The given information is

$$P(G \mid A) = \frac{1}{10{,}000}, \qquad P(G \mid M) = 0.2, \qquad P(A \mid G, M) = 0.5, \qquad P(A \mid G^c, M) = P(A) = 0.1.$$

The defense's number $P(G \mid A)$ ignores the one thing we know for certain: the wife *was* murdered.

The right quantity is $P(G \mid A, M)$. Using Bayes' rule with extra conditioning on M, and LOTP for the denominator:

$$P(G \mid A, M) = \frac{P(A \mid G, M)\, P(G \mid M)}{P(A \mid G, M)\, P(G \mid M) + P(A \mid G^c, M)\, P(G^c \mid M)}$$

$$= \frac{(0.5)(0.2)}{(0.5)(0.2) + (0.1)(0.8)} \qquad \text{plug in}$$

$$= \frac{0.10}{0.18} = \frac{5}{9} \approx 0.56.$$

The abuse history raises the probability of guilt from $P(G \mid M) = 0.2$ to about 0.56, so it is very relevant. This is also about 5,600 times bigger than the defense's 1/10,000. Notice we never used $P(G \mid A)$ anywhere: it describes the wrong population.

Sanity check with counts. Picture 1000 murdered wives:

- 200 were killed by their husbands, and half of those husbands, 100, were abusive;
- 800 were killed by someone else, and 10% of those husbands, 80, were abusive.

That's 180 abusive husbands in total, 100 of them guilty, so $P(G \mid A, M) = 100/180 = 5/9$.

Pitfall. Before using a conditional probability, list *everything* you know and check that all of it is to the right of the bar. A number that's correct but conditioned on the wrong information is still the wrong number.

Simpson's Paradox

Definition 2.6 (Simpson's Paradox)

Events A, B, C give a **Simpson's paradox** if

$$P(A \mid B, C) < P(A \mid B^c, C) \quad \text{and} \quad P(A \mid B, C^c) < P(A \mid B^c, C^c),$$

but

$$P(A \mid B) > P(A \mid B^c).$$

So B^c does better than B whether or not C happens, yet B does better once we stop separating by C.

Intuition. LOTP with extra conditioning writes each overall rate as a **weighted average** of the two subgroup rates:

$$P(A \mid B) = P(A \mid C, B)\, P(C \mid B) + P(A \mid C^c, B)\, P(C^c \mid B),$$

$$P(A \mid B^c) = P(A \mid C, B^c)\, P(C \mid B^c) + P(A \mid C^c, B^c)\, P(C^c \mid B^c).$$

Winning both subgroup comparisons doesn't guarantee winning the overall comparison, because the two averages can use very different weights. If B^c spends most of its time in the hard subgroup and B spends most of its time in the easy one, the pooled numbers can flip. The variable C that causes this is called a **confounder**: it's related both to which group you're in (B or B^c) and to the outcome (A).

Example 2.7 (Dr. Hibbert vs. Dr. Nick)

Two doctors each do two kinds of procedures: heart surgery (hard) and Band-Aid removal (easy). Their records are:

	Dr. Hibbert		**Dr. Nick**	
Procedure	Successes	Rate	Successes	Rate
Heart surgery	70/90	77.8%	2/10	20.0%
Band-Aid removal	10/10	100%	81/90	90.0%
Overall	80/100	80.0%	83/100	83.0%

Dr. Hibbert is better at both procedures, but Dr. Nick has the better overall rate. Explain this using LOTP.

Solution. Let A be a successful procedure, B that Dr. Nick is the surgeon (so B^c is Dr. Hibbert), and C that the procedure is heart surgery. The weights are each doctor's mix of procedures:

$$P(C \mid B) = \frac{10}{100} = 0.1, \qquad P(C \mid B^c) = \frac{90}{100} = 0.9.$$

Now each overall rate is a weighted average of that doctor's two subgroup rates:

$$\begin{aligned} P(A \mid B) &= P(A \mid C, B)\,P(C \mid B) + P(A \mid C^c, B)\,P(C^c \mid B) && \text{LOTP, extra cond. on } B \\ &= \frac{2}{10}(0.1) + \frac{81}{90}(0.9) = 0.02 + 0.81 = 0.83, \end{aligned}$$

$$\begin{aligned} P(A \mid B^c) &= P(A \mid C, B^c)\,P(C \mid B^c) + P(A \mid C^c, B^c)\,P(C^c \mid B^c) && \text{LOTP, extra cond. on } B^c \\ &= \frac{70}{90}(0.9) + \frac{10}{10}(0.1) = 0.70 + 0.10 = 0.80. \end{aligned}$$

Dr. Nick's average puts 90% of its weight on the easy procedure, and Dr. Hibbert's puts 90% on the hard one. That's the whole paradox. Hibbert's overall rate is lower because he takes on the risky cases, not because he's worse at anything.

Proposition 2.8 (Same weights means no paradox)

If $P(C \mid B) = P(C \mid B^c)$, then Simpson's paradox cannot happen: if B does worse than B^c given C and also given C^c, then $P(A \mid B) < P(A \mid B^c)$.

Proof. Let $w = P(C \mid B) = P(C \mid B^c)$, so $1 - w = P(C^c \mid B) = P(C^c \mid B^c)$. By LOTP with extra conditioning,

$$\begin{aligned} P(A \mid B) &= w\,P(A \mid C, B) + (1-w)\,P(A \mid C^c, B) && \text{LOTP} \\ &< w\,P(A \mid C, B^c) + (1-w)\,P(A \mid C^c, B^c) && \text{each rate is smaller, } w, 1-w \geq 0 \\ &= P(A \mid B^c). && \text{LOTP} \end{aligned}$$

The inequality is strict because w and $1 - w$ can't both be 0. □

So the paradox needs *both* ingredients: the confounder C has to affect the outcome, and the two groups have to have different amounts of C. Randomized experiments work by forcing $P(C \mid B) = P(C \mid B^c)$ for every possible C, which is exactly why they're immune to this.

Example 2.9 (Free throws by half)

Over a season, two basketball players have these free-throw records:

	Player X	**Player Y**
First half	$9/10 = 90\%$	$32/40 = 80\%$
Second half	$20/40 = 50\%$	$4/10 = 40\%$
Total	$29/50 = 58\%$	$36/50 = 72\%$

Identify A, B, C, and explain why Y wins overall.

Solution. Let A be a made shot, B that Player Y is shooting, and C that the shot is in the first half. X wins both halves ($90\% > 80\%$ and $50\% > 40\%$), so the subgroup inequalities hold. The weights are

$$P(C \mid B) = \frac{40}{50} = 0.8, \qquad P(C \mid B^c) = \frac{10}{50} = 0.2,$$

so Y takes most of his shots in the first half, when everyone shoots better. Checking with LOTP:

$$P(A \mid B) = (0.8)(0.8) + (0.4)(0.2) = 0.64 + 0.08 = 0.72,$$

$$P(A \mid B^c) = (0.9)(0.2) + (0.5)(0.8) = 0.18 + 0.40 = 0.58.$$

Since $0.72 > 0.58$, Y wins overall, even though X is the better shooter in each half.

Remark. Other real cases with the same structure: graduate admissions where women were admitted at higher rates within most departments but lower rates overall (they applied more to competitive departments), and mortality studies where one type of smoker looks healthier overall only because that group is younger on average. In each case, try to name A, B and C.

Pitfall. Pooled data hides who ended up in which subgroup. Whenever a third variable could be tied to both the grouping and the outcome, split the data by that variable before comparing.

2.9 Recap

Chapter 2 at a glance

- **Conditional probability.** For $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

  $P(\cdot \mid B)$ is itself a probability function, so every rule for probability also works with B added to the right of the bar.

- **Bayes' rule.** Flips the conditioning:

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}.$$

  Odds form: posterior odds = likelihood ratio $\times$ prior odds.

- **Law of total probability.** For a partition $A_1, \ldots, A_n$ of S,

$$P(B) = \sum_{i=1}^{n} P(B \mid A_i)\,P(A_i).$$

  Bayes' rule and LOTP are usually used together: LOTP gives the denominator.

- **Independence.** A and B are independent if $P(A \cap B) = P(A)\,P(B)$, i.e. learning B doesn't change the probability of A. Conditional independence given E does *not* imply independence, and independence does *not* imply conditional independence.

Common mistakes

1. Mixing up the prior $P(A)$ and the posterior $P(A \mid B)$.
2. **Prosecutor's fallacy:** confusing $P(A \mid B)$ with $P(B \mid A)$.
3. **Defense attorney's fallacy:** not conditioning on all the evidence.
4. **Simpson's paradox:** trusting pooled data without asking whether a confounder changes the weights.

Problem-solving strategies

- **Condition on what you wish you knew.** Pick the event that would make the problem easy if you knew it, then use LOTP over it and its complement.
- **First-step analysis.** For multi-stage problems, condition on what happens in the first step to get a recursive equation.
- **Update one piece of evidence at a time.** Today's posterior is tomorrow's prior:

$$P(A \mid B, C) = \frac{P(C \mid A, B)\, P(A \mid B)}{P(C \mid B)}.$$

Example 2.10 (Updating in two steps)

A coin is either fair or double-headed, each with probability $1/2$. We flip it on Monday and get Heads, and again on Tuesday and get Heads. Find the probability the coin is double-headed after each day, then check the final answer in one step.

Solution. Let D be the event the coin is double-headed, and H_1, H_2 the events of Heads on Monday and Tuesday. Given which coin it is, the flips are independent, with $P(H_i \mid D) = 1$ and $P(H_i \mid D^c) = 1/2$.

Monday.

$$P(D \mid H_1) = \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2}} = \frac{1/2}{3/4} = \frac{2}{3}.$$

Tuesday. Now 2/3 becomes the prior:

$$P(D \mid H_1, H_2) = \frac{P(H_2 \mid D, H_1)\, P(D \mid H_1)}{P(H_2 \mid D, H_1)\, P(D \mid H_1) + P(H_2 \mid D^c, H_1)\, P(D^c \mid H_1)} \quad \text{Bayes, extra cond. on } H_1$$

$$= \frac{1 \cdot \frac{2}{3}}{1 \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{3}} = \frac{2/3}{5/6} = \frac{4}{5}.$$

Check in one step. Using both flips at once with the original prior:

$$P(D \mid H_1, H_2) = \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2}} = \frac{1/2}{5/8} = \frac{4}{5}.$$

Same answer, as it has to be. Updating one step at a time and updating all at once always agree.

Chapter 3: Random Variables and their Distributions

3.1 Random Variables

Definition 3.1 (Random Variable)

Given an experiment with sample space S, a **random variable** (r.v.) is a function from the sample space to the real line:

$$X : S \to \mathbb{R}.$$

Intuition. An r.v. is a numerical summary of the outcome. The function X itself is completely fixed; all the randomness comes from the experiment picking an outcome $s \in S$. Once s is picked, X "crystallizes" into the number $X(s)$. Different r.v.s on the same S are just different ways of summarizing the same experiment.

Example 3.2 (Coin tosses)

Flip a fair coin twice, so $S = \{HH, HT, TH, TT\}$. Three r.v.s on this sample space:

- X = number of Heads: $X(HH) = 2$, $X(HT) = X(TH) = 1$, $X(TT) = 0$.
- Y = number of Tails: $Y = 2 - X$, meaning $Y(s) = 2 - X(s)$ for every $s \in S$.
- I = indicator that the first toss is Heads: $I(HH) = I(HT) = 1$, $I(TH) = I(TT) = 0$.

If we code $H = 1$ and $T = 0$, then $S = \{(1,1), (1,0), (0,1), (0,0)\}$ and each r.v. becomes an explicit formula:

$$X(s_1, s_2) = s_1 + s_2, \qquad Y(s_1, s_2) = 2 - s_1 - s_2, \qquad I(s_1, s_2) = s_1.$$

3.2 Distributions and Probability Mass Functions

Definition 3.3 (Discrete r.v. and Support)

An r.v. X is **discrete** if there is a finite or countably infinite list of values $a_1, a_2, \dots$ such that

$$P(X = a_j \text{ for some } j) = 1.$$

The **support** of X is the set of values x with $P(X = x) > 0$.

Example 3.4 (Supports)

For the two coin tosses, X has support $\{0, 1, 2\}$ (finite). If instead we flip until the first Heads and let N be the number of flips, N has support $\{1, 2, 3, \dots\}$ (countably infinite). Both are discrete.

Definition 3.5 (Probability Mass Function)

The **probability mass function** (PMF) of a discrete r.v. X is the function

$$p_X(x) = P(X = x).$$

It is positive on the support of X and 0 everywhere else.

Remark. $\{X = x\}$ is an *event*: it's shorthand for $\{s \in S : X(s) = x\}$. The PMF just takes the probability of that event for each x.

Example 3.6 (PMFs for the coin tosses)

Going back to Example 3.2, each of the 4 outcomes has probability 1/4, so

$$p_X(0) = \tfrac{1}{4}, \qquad p_X(1) = \tfrac{1}{2}, \qquad p_X(2) = \tfrac{1}{4},$$

$$p_Y(0) = \tfrac{1}{4}, \qquad p_Y(1) = \tfrac{1}{2}, \qquad p_Y(2) = \tfrac{1}{4},$$

$$p_I(0) = \tfrac{1}{2}, \qquad p_I(1) = \tfrac{1}{2}.$$

X and Y have the *same* PMF but are *different* r.v.s: they only agree on the outcomes HT and TH, where both equal 1.

Pitfall. An r.v. is not the same thing as its distribution. The distribution is a blueprint; different r.v.s can be built from the same blueprint, like X and Y above. "X and Y have the same distribution" does *not* mean $X = Y$.

Example 3.7 (Sum of two dice)

Roll two fair dice and let $T = X + Y$ be the total. All 36 ordered pairs are equally likely, so $P(T = t)$ is the number of pairs adding to t, divided by 36:

t	2	3	4	5	6	7	8	9	10	11	12
$36 \cdot P(T = t)$	1	2	3	4	5	6	5	4	3	2	1

The PMF is symmetric about 7: $P(T = t) = P(T = 14 - t)$. A compact formula is

$$P(T = t) = \frac{6 - |t - 7|}{36}, \qquad t = 2, 3, \ldots, 12.$$

Example 3.8 (Children in a household)

Pick a U.S. household uniformly at random and let X be the number of children in it. Then X is discrete with support contained in $\{0, 1, 2, \ldots\}$, and

$$P(X = x) = \frac{\text{number of households with } x \text{ children}}{\text{total number of households}}.$$

Theorem 3.9 (Valid PMFs)

Let X be discrete with support $x_1, x_2, \ldots$. A function p_X is a valid PMF if and only if

1. **Nonnegative:** $p_X(x) > 0$ if $x = x_j$ for some j, and $p_X(x) = 0$ otherwise;
2. **Sums to 1:** $\sum_{j=1}^{\infty} p_X(x_j) = 1$.

Intuition. The events $\{X = x_1\}, \{X = x_2\}, \ldots$ are disjoint and together they cover everything that can happen, so their probabilities have to add to 1. Nonnegativity is just the first axiom of probability.

Example 3.10 (Finding a normalizing constant)

Let $p(k) = c\left(\frac{1}{2}\right)^k$ for $k = 0, 1, 2, \ldots$ and 0 otherwise. Find c so that p is a valid PMF, then find $P(X \geq 3)$.

Solution. The values are clearly nonnegative for $c > 0$, so we just need the sum to be 1:

$$1 = \sum_{k=0}^{\infty} c\left(\tfrac{1}{2}\right)^k = c \cdot \frac{1}{1 - 1/2} \qquad \text{geometric series}$$

$$= 2c,$$

so $c = \frac{1}{2}$. Then using the complement,

$$P(X \geq 3) = 1 - \big(p(0) + p(1) + p(2)\big) = 1 - \left(\tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8}\right) = \frac{1}{8}.$$

3.3 Bernoulli and Binomial

Definition 3.11 (Bernoulli Distribution)

An r.v. X has the **Bernoulli distribution** with parameter p, written $X \sim \text{Bern}(p)$, if X only takes the values 0 and 1 and

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p,$$

where $0 < p < 1$. In one formula, $P(X = x) = p^x(1 - p)^{1-x}$ for $x \in \{0, 1\}$.

Example 3.12 (Lawyers)

Suppose 1 in 200 Canadian adults is a lawyer, and let X be 1 if a randomly chosen Canadian adult is a lawyer and 0 otherwise. Then $X \sim \text{Bern}(1/200)$:

$$P(X=1)=\frac{1}{200}, \qquad P(X=0)=\frac{199}{200}.$$

Definition 3.13 (Indicator Random Variable)

The **indicator** of an event A is the r.v. I_A that equals 1 if A happens and 0 otherwise. Every indicator is Bernoulli:

$$I_A \sim \text{Bern}\big(P(A)\big).$$

Example 3.14 (Indicator of a high roll)

Roll a fair die and let A be the event the roll is at least 5. Then $P(A)=2/6$, so $I_A \sim \text{Bern}(1/3)$. In Example 3.2, I is the indicator of "first toss is Heads", so $I \sim \text{Bern}(1/2)$.

Story 3.15 (Binomial Distribution)

Run n **independent** Bernoulli trials, each with the **same** success probability p. The number of successes X has the **Binomial distribution** with parameters n and p:

$$X \sim \text{Bin}(n,p).$$

Note that $\text{Bern}(p)$ is the same as $\text{Bin}(1,p)$.

Theorem 3.16 (Binomial PMF)

If $X \sim \text{Bin}(n,p)$ and $q=1-p$, then

$$P(X=k)=\binom{n}{k}p^k q^{n-k}, \qquad k=0,1,\dots,n,$$

and $P(X=k)=0$ otherwise.

Proof. Any *particular* sequence with k successes and $n-k$ failures, like $SSFSF\cdots$, has probability $p^k q^{n-k}$ by independence. There are $\binom{n}{k}$ such sequences, one for each choice of which k trials are

the successes. These sequences are disjoint, so we add them up to get $\binom{n}{k}p^k q^{n-k}$. This is a valid PMF by the binomial theorem:

$$\sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} = (p+q)^n = 1.$$

□

Example 3.17 (Guessing on a quiz)

A quiz has 10 multiple-choice questions with 4 options each. You guess every answer at random, independently. Find the probability of getting at least 7 right, and the probability of getting none right.

Solution. Each question is a success with probability $1/4$, independently, so the number correct is $X \sim \text{Bin}(10, 1/4)$. Then

$$\begin{aligned} P(X \geq 7) &= \sum_{k=7}^{10} \binom{10}{k} \left(\tfrac{1}{4}\right)^k \left(\tfrac{3}{4}\right)^{10-k} && \text{Binomial PMF} \\ &= \frac{\binom{10}{7}3^3 + \binom{10}{8}3^2 + \binom{10}{9}3 + \binom{10}{10}}{4^{10}} && \text{common denominator} \\ &= \frac{3240 + 405 + 30 + 1}{1{,}048{,}576} = \frac{3676}{1{,}048{,}576} \approx 0.0035. \end{aligned}$$

And

$$P(X = 0) = \left(\tfrac{3}{4}\right)^{10} \approx 0.056.$$

So guessing your way to 70% happens about 1 time in 285, while getting a zero happens about 1 time in 18.

Theorem 3.18 (Successes and Failures)

If $X \sim \text{Bin}(n, p)$ and $q = 1 - p$, then $n - X \sim \text{Bin}(n, q)$.

Proof. **Story proof.** If X counts successes, then $n - X$ counts failures. Relabelling "failure" as "success" turns each trial into a $\text{Bern}(q)$ trial, so $n - X \sim \text{Bin}(n, q)$.

Algebra proof. Let $Y = n - X$. For $k = 0, \ldots, n$,

$$\begin{aligned} P(Y = k) &= P(X = n - k) \\ &= \binom{n}{n-k} p^{n-k} q^k && \text{Binomial PMF} \\ &= \binom{n}{k} q^k p^{n-k}, && \tbinom{n}{n-k} = \tbinom{n}{k} \end{aligned}$$

which is the $\text{Bin}(n, q)$ PMF. □

Corollary 3.19 (Symmetry when $p = 1/2$)

If $X \sim \text{Bin}(n, 1/2)$, then $P(X = k) = P(X = n - k)$ for all k. In particular, if n is even, the PMF is symmetric about $n/2$:

$$P\big(X = \tfrac{n}{2} + j\big) = P\big(X = \tfrac{n}{2} - j\big) \quad \text{for all } j \geq 0.$$

Proof. With $p = q = 1/2$, Theorem 3.18 says X and $n - X$ are both $\text{Bin}(n, 1/2)$, so $P(X = k) = P(n - X = k) = P(X = n - k)$. Plugging in $k = n/2 + j$ gives the second statement. □

Example 3.20 (Checking with the coin tosses)

In Example 3.2, $X \sim \text{Bin}(2, 1/2)$ and $Y = 2 - X \sim \text{Bin}(2, 1/2)$ as well, which matches the PMFs we found. The PMF $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ is symmetric about $1 = n/2$.

3.4 Hypergeometric

Story 3.21 (Hypergeometric Distribution)

An urn has w white and b black balls. Draw n balls **without replacement**, so that all $\binom{w+b}{n}$ subsets of size n are equally likely. The number of white balls in the sample has the **Hypergeometric distribution**:

$$X \sim \text{HGeom}(w, b, n).$$

Theorem 3.22 (Hypergeometric PMF)

If $X \sim \text{HGeom}(w, b, n)$, then

$$P(X = k) = \frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}}$$

for integers k with $0 \le k \le w$ and $0 \le n - k \le b$, and $P(X = k) = 0$ otherwise.

Proof. This is the naive definition of probability. The denominator counts all equally likely samples of size n. For the numerator, choose which k of the w white balls are in the sample, and which $n - k$ of the b black balls are, and multiply by the multiplication rule. □

Remark. The range for k is the same as $\max(0,\, n-b) \le k \le \min(n,\, w)$: you can't get more white balls than you drew or than exist, and if $n > b$ you're forced to get at least $n - b$ white ones.

Example 3.23 (Aces in a poker hand)

Deal a 5-card hand from a shuffled 52-card deck. Find the probability of exactly 3 aces.

Solution. Aces are the "white balls" and non-aces the "black balls", so the number of aces is $X \sim \text{HGeom}(4, 48, 5)$. Then

$$P(X = 3) = \frac{\binom{4}{3}\binom{48}{2}}{\binom{52}{5}} \qquad \text{HGeom PMF}$$

$$= \frac{4 \cdot 1128}{2{,}598{,}960} \approx 0.00174.$$

Example 3.24 (Capture–recapture)

A forest has N elk. We catch m of them, tag them, and release them. Later we catch a new random sample of n elk. What is the distribution of the number of tagged elk in the new sample?

Solution. Tagged elk are white balls ($w = m$), untagged elk are black balls ($b = N - m$), and the second catch is a sample of size n without replacement. So

$$X \sim \text{HGeom}(m,\, N - m,\, n).$$

This is how ecologists estimate N: about m/N of the forest is tagged, so we expect about $n \cdot m/N$ tagged elk in the second catch. Solving $X \approx nm/N$ for N gives the estimate $N \approx nm/X$.

Theorem 3.25 (Swapping the Two Tags)

The $\text{HGeom}(w, b, n)$ and $\text{HGeom}(n,\, w + b - n,\, w)$ distributions are the same.

Proof. **Story proof.** Every ball carries two tags: white or black, and sampled or not sampled. X counts the balls that are both white *and* sampled. We can count those by treating "sampled" as the colour (n sampled balls, $w + b - n$ not) and "white" as the sample (a random set of w balls). That's the $\text{HGeom}(n, w + b - n, w)$ story.

Algebra proof. Expanding both PMFs into factorials gives the same thing:

$$\frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}} = \frac{w!\; b!\; n!\; (w+b-n)!}{k!\; (w+b)!\; (w-k)!\; (n-k)!\; (b-n+k)!} = \frac{\binom{n}{k}\binom{w+b-n}{w-k}}{\binom{w+b}{w}}.$$

□

Binomial vs. Hypergeometric

Both count successes in n draws. The difference is whether the draws are independent.

- **Binomial:** draws *with* replacement (or independent trials). Each draw has the same success probability p, no matter what happened before.
- **Hypergeometric:** draws *without* replacement. Each draw changes what's left in the urn, so the draws are dependent.

Quick test: if drawing a success makes the next success less likely, it's Hypergeometric.

3.5 Discrete Uniform

Story 3.26 (Discrete Uniform Distribution)

Let C be a finite, nonempty set of numbers. Pick one of them uniformly at random, so every element is equally likely. The number picked has the **Discrete Uniform distribution**, $X \sim \text{DUnif}(C)$, with PMF

$$P(X = x) = \frac{1}{|C|} \quad \text{for } x \in C,$$

and 0 otherwise. For any $A \subseteq C$,

$$P(X \in A) = \frac{|A|}{|C|}.$$

Intuition. This is just the naive definition of probability, written as an r.v. Counting favourable outcomes over total outcomes always gives a Discrete Uniform somewhere underneath.

Example 3.27 (Slips of paper)

A hat has 100 slips numbered $1, \ldots, 100$, and we draw 5. Compare drawing *with* and *without* replacement for: (a) the number of slips with value at least 80; (b) the value of the jth draw; (c) the chance that slip 100 shows up.

Solution. There are 21 slips with value at least 80 (namely $80, \ldots, 100$).

With replacement. The draws are independent and each is $\text{DUnif}(\{1, \ldots, 100\})$.

(a) Each draw is ≥ 80 with probability 0.21, independently, so $Y \sim \text{Bin}(5, 0.21)$.

(b) $X_j \sim \text{DUnif}(\{1, \ldots, 100\})$.

(c) Using the complement,

$$P(100 \text{ drawn at least once}) = 1 - \left(\tfrac{99}{100}\right)^5 \approx 0.0490.$$

Without replacement. All $\binom{100}{5}$ sets of slips are equally likely.

(a) $Y \sim \text{HGeom}(21, 79, 5)$.

(b) By symmetry, the jth draw is equally likely to be any of the 100 slips, so $Y_j \sim \text{DUnif}(\{1, \dots, 100\})$ *unconditionally.* (Given earlier draws, it isn't.)

(c) Count the samples that contain slip 100:

$$P(100 \in \text{sample}) = \frac{\binom{1}{1}\binom{99}{4}}{\binom{100}{5}} = \frac{5}{100} = 0.05.$$

The two answers to (c) are close because with only 5 draws out of 100, repeats are rare, so with and without replacement barely differ. Section 3.9 makes this precise.

3.6 Cumulative Distribution Functions

Definition 3.28 (Cumulative Distribution Function)

The **cumulative distribution function** (CDF) of an r.v. X is

$$F_X(x) = P(X \le x), \qquad x \in \mathbb{R}.$$

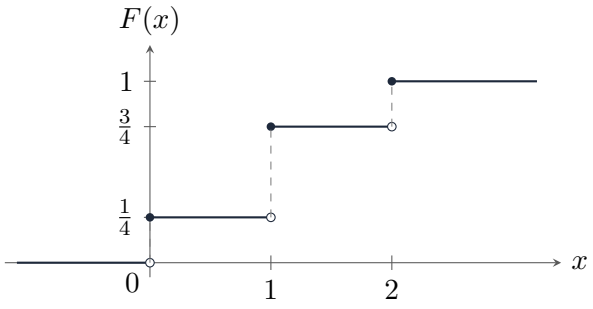
Intuition. The PMF only works for discrete r.v.s, but *every* r.v. has a CDF. For a discrete r.v., the CDF is a staircase: flat between values in the support, and jumping up at each support point by exactly the PMF there.

Example 3.29 (CDF of the number of Heads)

Find and sketch the CDF of $X \sim \text{Bin}(2, 1/2)$, the number of Heads in two tosses.

Solution. Add up the PMF $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ at $0, 1, 2$ over all support values $\le x$:

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{4}, & 0 \le x < 1, \\ \frac{3}{4}, & 1 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$



The jumps at $0, 1, 2$ have sizes $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$, which are exactly the PMF values. The filled dots show that F takes the *upper* value at each jump, which is right-continuity.

Going between the PMF and the CDF (discrete X)

$$F(x) = \sum_{t \le x} P(X = t) \qquad \text{PMF} \to \text{CDF: add up}$$

$$P(X = x) = F(x) - \lim_{t \to x^-} F(t) \qquad \text{CDF} \to \text{PMF: size of the jump}$$

$$P(a < X \le b) = F(b) - F(a) \qquad \text{probability of an interval}$$

Watch the endpoints: $<$ vs. $\le$ matters for discrete r.v.s.

Example 3.30 (An interval for the dice total)

For the dice total T from Example 3.7, find $P(4 < T \le 7)$ using the CDF.

Solution. From the table, $F(4) = \frac{1+2+3}{36} = \frac{6}{36}$ and $F(7) = \frac{1+2+3+4+5+6}{36} = \frac{21}{36}$. So

$$P(4 < T \le 7) = F(7) - F(4) = \frac{21 - 6}{36} = \frac{15}{36}.$$

Check: $P(T = 5) + P(T = 6) + P(T = 7) = \frac{4+5+6}{36} = \frac{15}{36}$.

Theorem 3.31 (Valid CDFs)

Any CDF F has these properties, and any function with these properties is the CDF of some r.v.:

1. **Increasing:** if $x_1 \leq x_2$, then $F(x_1) \leq F(x_2)$.
2. **Right-continuous:** $F(x) = \lim_{t \to x^+} F(t)$ for all x.
3. **Limits:** $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$.

Intuition. (1) As x grows, the event $\{X \leq x\}$ only gets bigger. (2) F is continuous from the right because $\{X \leq x\}$ includes the endpoint x, so approaching from the right you already "have" the jump. (3) Far to the left nothing is $\leq x$; far to the right everything is.

3.7 Functions of Random Variables

Definition 3.32 (Function of an r.v.)

For an r.v. X and a function $g : \mathbb{R} \to \mathbb{R}$, $g(X)$ is the r.v. that maps each outcome s to $g(X(s))$. More generally, for r.v.s X and Y, $g(X, Y)$ is the r.v. that maps s to $g(X(s), Y(s))$.

Example 3.33 (Functions on the coin tosses)

In Example 3.2, $Y = 2 - X$ is $g(X)$ with $g(x) = 2 - x$. And X^2 is the r.v. with $X^2(HH) = 4$, $X^2(HT) = X^2(TH) = 1$, $X^2(TT) = 0$. "First apply X, then apply g."

Theorem 3.34 (PMF of $g(X)$)

Let X be discrete and $g : \mathbb{R} \to \mathbb{R}$. Then the support of $g(X)$ is the set of all $g(x)$ with x in the support of X, and

$$P\big(g(X) = y\big) = \sum_{x\,:\,g(x)=y} P(X = x).$$

If g is one-to-one, there's only one term: $P\big(g(X) = g(x)\big) = P(X = x)$.

Intuition. To find $P(g(X) = y)$, collect every value x that g sends to y and add up their probabilities. If g is one-to-one, nothing gets merged and the probabilities just get relabelled.

Example 3.35 (Random walk)

A particle starts at 0 and takes n independent steps, each $+1$ or -1 with probability $1/2$. Let Y be its position after n steps. Find the PMF of Y.

Solution. Let X be the number of $+1$ steps. Each step is an independent Bern$(1/2)$ trial, so $X \sim \text{Bin}(n, 1/2)$. The other $n - X$ steps are -1, so

$$Y = X(+1) + (n - X)(-1) = 2X - n.$$

$g(x) = 2x - n$ is one-to-one, and $2x - n = k$ means $x = \frac{n+k}{2}$. So for $k \in \{-n, -n+2, \ldots, n-2, n\}$,

$$P(Y = k) = P\left(X = \frac{n+k}{2}\right) = \binom{n}{\frac{n+k}{2}} \left(\frac{1}{2}\right)^n.$$

Y always has the same parity as n, which is why k goes up in steps of 2.

Example 3.36 (Distance from the origin)

Continuing Example 3.35 with n even, let $D = |Y|$. Find the PMF of D.

Solution. Now $g(y) = |y|$ is *not* one-to-one: both k and $-k$ map to k. So we have to merge them. For $k = 0$ there's only one preimage:

$$P(D=0) = P(Y=0) = \binom{n}{n/2}\left(\frac{1}{2}\right)^n.$$

For $k = 2, 4, \ldots, n$,

$$\begin{aligned} P(D=k) &= P(Y=k) + P(Y=-k) && \text{PMF of } g(X) \\ &= 2\binom{n}{\frac{n+k}{2}}\left(\frac{1}{2}\right)^n. && \text{symmetry of } Y \end{aligned}$$

Example 3.37 (Max of two dice)

Roll two fair dice X and Y and let $M = \max(X, Y)$. Find the PMF of M.

Solution. Going through the CDF is easiest, since "the max is at most k" just means "both are at most k":

$$\begin{aligned} P(M \le k) &= P(X \le k, Y \le k) && \max \le k \text{ iff both} \le k \\ &= P(X \le k)\, P(Y \le k) && \text{independent dice} \\ &= \left(\frac{k}{6}\right)^2, \qquad k = 1, \ldots, 6. \end{aligned}$$

Then the PMF is the jump in the CDF:

$$P(M=k) = P(M \le k) - P(M \le k-1) = \frac{k^2 - (k-1)^2}{36} = \frac{2k-1}{36}.$$

k	1	2	3	4	5	6
$36 \cdot P(M=k)$	1	3	5	7	9	11

Check: $1 + 3 + 5 + 7 + 9 + 11 = 36$.

Pitfall. Don't do algebra on PMFs as if they were the r.v.s. If X has PMF p_X, the PMF of $2X$ is *not* $2p_X(x)$, and the PMF of $X + Y$ is *not* $p_X + p_Y$. Always go back to events: $P(2X = y) = P(X = y/2)$.

3.8 Independence of Random Variables

Definition 3.38 (Independence of r.v.s)

X and Y are **independent** if

$$P(X \leq x, Y \leq y) = P(X \leq x)\, P(Y \leq y) \quad \text{for all } x, y \in \mathbb{R}.$$

For discrete r.v.s this is equivalent to

$$P(X = x, Y = y) = P(X = x)\, P(Y = y) \quad \text{for all } x, y.$$

$X_1, \ldots, X_n$ are independent if $P(X_1 \leq x_1, \ldots, X_n \leq x_n) = P(X_1 \leq x_1) \cdots P(X_n \leq x_n)$ for all $x_1, \ldots, x_n$.

Intuition. Knowing the value of X tells you nothing about Y. Notice it has to hold for *all* x and y, so a single pair of values that breaks the product rule is enough to show dependence. Also, if X and Y are independent, then any function of X is independent of any function of Y (e.g. X^2 and e^Y).

Example 3.39 (A die and the total)

Roll two fair dice X and Y, and let $T = X + Y$. Are X and T independent?

Solution. No. One pair of values is enough:

$$P(X = 1, T = 12) = 0, \qquad \text{but} \qquad P(X = 1)\, P(T = 12) = \frac{1}{6} \cdot \frac{1}{36} \neq 0.$$

If the first die is 1, the total can't be 12, so X gives information about T.

Definition 3.40 (i.i.d.)

R.v.s that are independent *and* all have the same distribution are called **independent and identically distributed**, or **i.i.d.**

Example 3.41 (The two properties are separate)

"Independent" and "identically distributed" have nothing to do with each other. All four combinations happen:

	Identically distributed	**Not identically distributed**
Independent	two separate die rolls	a die roll and a separate coin flip
Dependent	# Heads and # Tails in n flips	a die roll X and X^2

In the bottom-left case, both are $\text{Bin}(n, 1/2)$, but knowing one tells you the other exactly.

Theorem 3.42 (Binomial as a Sum of Bernoullis)

If $X \sim \text{Bin}(n, p)$, we can write $X = X_1 + \cdots + X_n$ where $X_1, \ldots, X_n$ are i.i.d. $\text{Bern}(p)$.

Proof. Let X_i be the indicator that trial i is a success. Adding up the indicators counts the successes. □

Theorem 3.43 (Sum of Independent Binomials)

If $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ are independent (same p), then

$$X + Y \sim \text{Bin}(n + m, p).$$

Proof. **Story proof.** X counts successes in n independent $\text{Bern}(p)$ trials and Y counts successes in m more, independent of the first n. Together that's $n + m$ independent $\text{Bern}(p)$ trials, and $X + Y$ counts all the successes.

LOTP proof. Condition on X and let $q = 1 - p$. Terms with $j > n$ or $k - j > m$ are 0 automatically, so we can sum over $j = 0, \ldots, k$:

$$
\begin{aligned}
P(X+Y=k) &= \sum_{j=0}^{k} P(X+Y=k \mid X=j)\, P(X=j) && \text{LOTP} \\
&= \sum_{j=0}^{k} P(Y=k-j)\, P(X=j) && \text{independence} \\
&= \sum_{j=0}^{k} \binom{m}{k-j} p^{k-j} q^{m-k+j} \binom{n}{j} p^{j} q^{n-j} && \text{Binomial PMFs} \\
&= p^{k} q^{n+m-k} \sum_{j=0}^{k} \binom{n}{j}\binom{m}{k-j} && \text{factor out} \\
&= \binom{n+m}{k} p^{k} q^{n+m-k}. && \text{Vandermonde}
\end{aligned}
$$

□

Pitfall. The p's have to match. If $X \sim \text{Bin}(n, p_1)$ and $Y \sim \text{Bin}(m, p_2)$ with $p_1 \neq p_2$, then $X+Y$ is *not* Binomial, because the trials no longer all have the same success probability.

Definition 3.44 (Conditional Independence)

X and Y are **conditionally independent given** Z if for all x, y and all z in the support of Z,

$$P(X \leq x, Y \leq y \mid Z = z) = P(X \leq x \mid Z = z)\, P(Y \leq y \mid Z = z).$$

For discrete r.v.s, the same thing with $=$ in place of $\leq$ is equivalent. The function $x \mapsto P(X = x \mid Z = z)$ is called the **conditional PMF** of X given $Z = z$.

Remark. Independence does *not* imply conditional independence, and conditional independence does *not* imply independence. The next two examples show one direction each.

Example 3.45 (Matching pennies)

Players A and B each flip a fair penny, independently. Let $X = 1$ if A gets Heads and $X = -1$ otherwise, and define Y the same way for B. Let $Z = XY$, so $Z = 1$ when the pennies match. Show X and Y are independent but not conditionally independent given Z.

Solution. X and Y are independent by assumption. Given $Z = 1$ the pennies match, so $X = Y$. Each of $X = Y = 1$ and $X = Y = -1$ is equally likely given $Z = 1$, so

$$P(X = 1, Y = 1 \mid Z = 1) = \frac{1}{2}, \qquad \text{but} \qquad P(X = 1 \mid Z = 1)\,P(Y = 1 \mid Z = 1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Once we know Z, learning X tells us Y exactly. Independent, but conditionally dependent.

Example 3.46 (Mystery opponent)

You play two games of tennis against one of two identical twins, but you don't know which. You win each game with probability 1/2 against Twin 1 and 3/4 against Twin 2, and each twin is equally likely to be your opponent. Let Z indicate that it's Twin 1, and let X, Y indicate wins in games 1 and 2. Show X and Y are conditionally independent given Z, but not independent.

Solution. Given $Z = 1$, X and Y are i.i.d. Bern(1/2); given $Z = 0$, they're i.i.d. Bern(3/4). So they're conditionally independent given Z. Unconditionally, by LOTP,

$$P(X = 1) = \tfrac{1}{2} \cdot \tfrac{1}{2} + \tfrac{1}{2} \cdot \tfrac{3}{4} = \tfrac{5}{8}, \qquad \text{LOTP on } Z$$

$$P(X = 1, Y = 1) = \tfrac{1}{2} \cdot \left(\tfrac{1}{2}\right)^2 + \tfrac{1}{2} \cdot \left(\tfrac{3}{4}\right)^2 = \tfrac{1}{8} + \tfrac{9}{32} = \tfrac{13}{32}. \qquad \text{LOTP on } Z$$

So

$$P(Y = 1 \mid X = 1) = \frac{13/32}{5/8} = \frac{13}{20} = 0.65 \;>\; 0.625 = \frac{5}{8} = P(Y = 1).$$

Winning game 1 makes it more likely you're facing the weaker twin (Twin 2), which makes winning game 2 more likely. So X and Y are dependent.

3.9 Connections between Binomial and Hypergeometric

The two distributions are linked both ways: conditioning turns a Binomial into a Hypergeometric, and a limit turns a Hypergeometric into a Binomial.

Theorem 3.47 (Binomial to Hypergeometric by Conditioning)

If $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ are independent, then given $X + Y = r$,

$$X \mid (X + Y = r) \;\sim\; \text{HGeom}(n, m, r).$$

Proof. Let $q = 1 - p$. By Bayes' rule,

$$\begin{aligned} P(X = x \mid X + Y = r) &= \frac{P(X + Y = r \mid X = x)\, P(X = x)}{P(X + Y = r)} && \text{Bayes' rule} \\ &= \frac{P(Y = r - x)\, P(X = x)}{P(X + Y = r)} && \text{independence} \\ &= \frac{\binom{m}{r-x} p^{r-x} q^{m-r+x} \cdot \binom{n}{x} p^x q^{n-x}}{\binom{n+m}{r} p^r q^{n+m-r}} && \text{Thm 3.43 for the bottom} \\ &= \frac{\binom{n}{x}\binom{m}{r-x}}{\binom{n+m}{r}}. && \text{powers of } p, q \text{ cancel} \end{aligned}$$

□

Intuition. Think of the $n + m$ trials as balls: the first n are "white", the last m are "black". Knowing there were r successes, every set of r trials is equally likely to be the successful ones (all those sequences have the same probability $p^r q^{n+m-r}$). So X is the number of white balls in a random sample of r, which is $\text{HGeom}(n, m, r)$. Notice p disappears completely.

Example 3.48 (Where did the Heads land?)

A coin with unknown bias p is flipped 10 times and lands Heads exactly 4 times. What's the probability that exactly 2 of those Heads were in the first 6 flips?

Solution. Let $X \sim \text{Bin}(6, p)$ be Heads in the first 6 flips and $Y \sim \text{Bin}(4, p)$ Heads in the last 4, independent. Given $X + Y = 4$, Theorem 3.47 gives $X \sim \text{HGeom}(6, 4, 4)$, so

$$P(X = 2 \mid X + Y = 4) = \frac{\binom{6}{2}\binom{4}{2}}{\binom{10}{4}} = \frac{15 \cdot 6}{210} = \frac{3}{7},$$

whatever p is.

Theorem 3.49 (Hypergeometric to Binomial as a Limit)

Let $X \sim \text{HGeom}(w, b, n)$ and $N = w + b$. If $N \to \infty$ with $p = w/N$ held fixed, then for each k,

$$P(X = k) \to \binom{n}{k} p^k q^{n-k}, \qquad q = 1 - p,$$

the $\text{Bin}(n, p)$ PMF.

Proof. Using Theorem 3.25 and writing out the factorials,

$$\begin{aligned} P(X = k) &= \binom{n}{k} \frac{\binom{w+b-n}{w-k}}{\binom{w+b}{w}} && \text{Thm 3.25} \\ &= \binom{n}{k} \frac{w(w-1)\cdots(w-k+1) \;\cdot\; b(b-1)\cdots(b-n+k+1)}{N(N-1)\cdots(N-n+1)} && \text{cancel factorials} \\ &= \binom{n}{k} \frac{p\left(p-\frac{1}{N}\right)\cdots\left(p-\frac{k-1}{N}\right)\cdot q\left(q-\frac{1}{N}\right)\cdots\left(q-\frac{n-k-1}{N}\right)}{\left(1-\frac{1}{N}\right)\left(1-\frac{2}{N}\right)\cdots\left(1-\frac{n-1}{N}\right)}. && \text{divide by } N^n \end{aligned}$$

As $N \to \infty$ every $\frac{j}{N} \to 0$, so the top goes to $p^k q^{n-k}$ and the bottom goes to 1. □

Intuition. If the urn is huge compared to the sample, taking a few balls out barely changes the proportion of white balls, so drawing without replacement is basically the same as drawing with replacement. Rule of thumb: if N is much bigger than n, use $\text{Bin}(n, w/N)$ as an approximation.

Example 3.50 (How big is "big enough"?)

An urn is 20% white and we draw $n = 5$ balls without replacement. Compare $P(X = 1)$ to the $\text{Bin}(5, 0.2)$ value as the urn grows.

Solution. The Binomial value is $\binom{5}{1}(0.2)(0.8)^4 = 0.4096$. The Hypergeometric values $\binom{w}{1}\binom{b}{4}/\binom{N}{5}$ with $w = 0.2N$ are:

Urn size N	20	50	500	5000	$\text{Bin}(5, 0.2)$
$P(X = 1)$	0.4696	0.4313	0.4117	0.4098	0.4096

With $N = 500$ (sample is 1% of the urn) the approximation is already off by less than 0.003.

3.10 Recap

Chapter 3 at a glance

- An **r.v.** is a function $X : S \to \mathbb{R}$. Its **distribution** says how probability is spread over its values, and can be given by a PMF (discrete only), a CDF (any r.v.), or a story.
- **Valid PMF:** nonnegative, sums to 1. **Valid CDF:** increasing, right-continuous, 0 at $-\infty$, 1 at $+\infty$.
- An r.v. is *not* its distribution. Different r.v.s can share one distribution.
- A **function of an r.v.** is an r.v. To get the PMF of $g(X)$, add up $P(X = x)$ over all x with $g(x) = y$.
- **Independent** and **identically distributed** are separate ideas. Independence doesn't imply conditional independence, or vice versa.

Named distributions so far

Name	Story	PMF	Support
Bern(p)	indicator of one success	$p^k q^{1-k}$	$\{0, 1\}$
Bin(n, p)	# successes, n indep. trials	$\binom{n}{k} p^k q^{n-k}$	$\{0, \dots, n\}$
HGeom(w, b, n)	# white, n draws w/o repl.	$\frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}}$	$\max(0, n-b) \le k \le \min(n, w)$
DUnif(C)	uniform pick from finite C	$\frac{1}{\lvert C\rvert}$	C

Common mistakes

1. Treating "same distribution" as "same r.v."
2. Using Binomial when draws are without replacement (that's Hypergeometric), or when the success probability changes between trials.
3. Adding two Binomials with different p and calling the result Binomial.
4. Doing algebra on PMFs instead of on events.
5. Forgetting to merge values when g isn't one-to-one.

Chapter 4: Expectation

4.1 Definition of Expectation

Definition 4.1 (Expectation of a Discrete r.v.)

The **expected value** (also called the **expectation** or **mean**) of a discrete r.v. X with possible values $x_1, x_2, \ldots$ is

$$\mathrm{E}(X) = \sum_j x_j \, P(X = x_j),$$

as long as the sum converges absolutely, i.e. $\sum_j |x_j| \, P(X = x_j) < \infty$. Otherwise we say $\mathrm{E}(X)$ does not exist (or is infinite).

Intuition. $\mathrm{E}(X)$ is a weighted average of the values of X, weighted by how likely each value is. Two ways to picture it: (1) if you repeat the experiment many times, the average of the X values you see settles near $\mathrm{E}(X)$; (2) if you put weight $P(X = x)$ at each point x on a number line, $\mathrm{E}(X)$ is where it balances.

Example 4.2 (Fair die)

Let X be a fair die roll. Then

$$\mathrm{E}(X) = \sum_{j=1}^{6} j \cdot \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

The mean doesn't have to be a value X can actually take.

Example 4.3 (Bernoulli expectation)

If $X \sim \mathrm{Bern}(p)$, then

$$\mathrm{E}(X) = 1 \cdot p + 0 \cdot (1-p) = p.$$

Proposition 4.4 (Expectation Depends Only on the Distribution)

If X and Y are discrete r.v.s with the same distribution, then $\mathrm{E}(X) = \mathrm{E}(Y)$ (if either exists).

Proof. $\mathrm{E}(X)$ is computed only from the values of X and their probabilities, i.e. from the PMF. Same PMF, same sum. □

Pitfall. The converse is false. $\mathrm{E}(X)$ is a one-number summary. For example, $X \sim \mathrm{Bern}(1/2)$ and Y that is always $1/2$ both have mean $1/2$, but completely different distributions.

4.2 Linearity of Expectation

Theorem 4.5 (Linearity of Expectation)

For any r.v.s X, Y and any constant c,

$$\mathrm{E}(X+Y) = \mathrm{E}(X) + \mathrm{E}(Y), \qquad \mathrm{E}(cX) = c\,\mathrm{E}(X).$$

This holds **whether or not** X and Y are independent.

Proof. For a finite or countable sample space, we can write any expectation as a sum over outcomes, $\mathrm{E}(X) = \sum_{s\in S} X(s)\,P(\{s\})$. (Grouping the outcomes by the value of X gives back the usual definition.) Then

$$\begin{aligned}
\mathrm{E}(X) + \mathrm{E}(Y) &= \sum_{s\in S} X(s)\,P(\{s\}) + \sum_{s\in S} Y(s)\,P(\{s\}) && \text{sum over outcomes}\\
&= \sum_{s\in S} \big(X(s) + Y(s)\big)\,P(\{s\}) && \text{combine sums}\\
&= \sum_{s\in S} (X+Y)(s)\,P(\{s\}) = \mathrm{E}(X+Y).
\end{aligned}$$

The proof of $\mathrm{E}(cX) = c\,\mathrm{E}(X)$ is the same, pulling c out of the sum. □

Intuition. Linearity works "pebble by pebble": each outcome s contributes $X(s)+Y(s)$ to the left side and $X(s)$ and $Y(s)$ separately to the right side. Dependence between X and Y never enters, because we never needed to know how X and Y relate, only their values at each outcome.

Example 4.6 (Binomial expectation)

Let $X \sim \text{Bin}(n,p)$. Write $X = I_1 + \cdots + I_n$, where I_j is the indicator of success on trial j, so $I_j \sim \text{Bern}(p)$. By linearity,

$$\text{E}(X) = \sum_{j=1}^{n} \text{E}(I_j) = \sum_{j=1}^{n} p = np.$$

Compare this to computing $\sum_k k\binom{n}{k}p^k q^{n-k}$ directly. Linearity saves a lot of algebra.

Example 4.7 (Hypergeometric expectation)

Let $X \sim \text{HGeom}(w,b,n)$. Find $\text{E}(X)$.

Solution. Write $X = I_1 + \cdots + I_n$, where I_j indicates that the jth ball drawn is white. By symmetry, the jth ball is equally likely to be any of the $w+b$ balls, so $I_j \sim \text{Bern}\left(\frac{w}{w+b}\right)$. The I_j are *dependent* (drawing without replacement), but linearity doesn't care:

$$\text{E}(X) = \sum_{j=1}^{n} \text{E}(I_j) = \frac{nw}{w+b}.$$

Same as a Binomial with $p = \frac{w}{w+b}$. The dependence affects the spread, not the mean.

Proposition 4.8 (Monotonicity of Expectation)

If $X \geq Y$ with probability 1, then $\text{E}(X) \geq \text{E}(Y)$, with equality if and only if $X = Y$ with probability 1.

Proof. Let $Z = X - Y$, which is nonnegative with probability 1. Then $\text{E}(Z)$ is a sum of nonnegative terms, so $\text{E}(Z) \geq 0$. By linearity, $\text{E}(X) - \text{E}(Y) = \text{E}(Z) \geq 0$. If $\text{E}(Z) = 0$, every term $z\,P(Z=z)$ must be 0, so $P(Z=0) = 1$. □

4.3 Geometric and Negative Binomial

Story 4.9 (Geometric Distribution)

Run independent Bern(p) trials. The number of **failures before the first success** has the **Geometric distribution**, $X \sim \text{Geom}(p)$, with support $\{0, 1, 2, \dots\}$.

Theorem 4.10 (Geometric PMF and CDF)

If $X \sim \text{Geom}(p)$ and $q = 1 - p$, then

$$P(X = k) = q^k p, \qquad k = 0, 1, 2, \dots,$$

and the CDF is

$$F(x) = \begin{cases} 1 - q^{\lfloor x \rfloor + 1}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Proof. $X = k$ means the first k trials fail and trial $k + 1$ succeeds: probability $q^k p$ by independence. For the CDF with integer $n \geq 0$, $P(X > n)$ is the chance the first $n + 1$ trials all fail, which is q^{n+1}. So $P(X \leq n) = 1 - q^{n+1}$, and between integers the CDF is flat. □

Theorem 4.11 (Geometric Expectation)

If $X \sim \text{Geom}(p)$, then $\text{E}(X) = \dfrac{q}{p}$.

Proof. Start from the geometric series and differentiate both sides with respect to q:

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q} \quad \Longrightarrow \quad \sum_{k=1}^{\infty} k q^{k-1} = \frac{1}{(1-q)^2} = \frac{1}{p^2}. \qquad \tfrac{d}{dq} \text{ both sides}$$

Then

$$\begin{aligned}
\mathrm{E}(X) &= \sum_{k=0}^{\infty} k\, q^k p && \text{definition} \\
&= pq \sum_{k=1}^{\infty} k\, q^{k-1} && \text{pull out } pq \\
&= pq \cdot \frac{1}{p^2} = \frac{q}{p}.
\end{aligned}$$

□

Story 4.12 (First Success Distribution)

Run independent Bern(p) trials. The number of trials **up to and including** the first success has the **First Success distribution**, $Y \sim \mathrm{FS}(p)$, with support $\{1, 2, 3, \dots\}$ and PMF

$$P(Y = k) = q^{k-1} p, \qquad k = 1, 2, \dots$$

If $X \sim \mathrm{Geom}(p)$ then $X + 1 \sim \mathrm{FS}(p)$, so

$$\mathrm{E}(Y) = \mathrm{E}(X) + 1 = \frac{q}{p} + 1 = \frac{1}{p}.$$

Pitfall. Different books define "Geometric" differently. Here, Geom(p) counts *failures* (starts at 0) and FS(p) counts *trials* (starts at 1). Always check which one a problem means.

Example 4.13 (Rolling for a six)

You roll a fair die until you get a 6. On average, how many rolls does it take, and how many non-sixes do you see first?

Solution. The number of rolls is FS(1/6), so on average $1/(1/6) = 6$ rolls. The number of non-sixes before the first six is Geom(1/6), with mean $\frac{5/6}{1/6} = 5$. These agree: 6 rolls is 5 failures plus the success.

Story 4.14 (Negative Binomial Distribution)

Run independent Bern(p) trials. The number of **failures before the rth success** has the **Negative Binomial distribution**, $X \sim \text{NBin}(r, p)$. Note $\text{NBin}(1, p) = \text{Geom}(p)$.

Theorem 4.15 (Negative Binomial PMF)

If $X \sim \text{NBin}(r, p)$, then

$$P(X = n) = \binom{n + r - 1}{r - 1} p^r q^n, \qquad n = 0, 1, 2, \ldots$$

Proof. $X = n$ means there are $n + r$ trials total, and the last one is the rth success. Any specific such sequence has probability $p^r q^n$. The last trial is fixed as a success, so we only choose where the other $r - 1$ successes go among the first $n + r - 1$ trials: $\binom{n+r-1}{r-1}$ ways. □

Theorem 4.16 (Negative Binomial as a Sum of Geometrics)

If $X \sim \text{NBin}(r, p)$, then $X = X_1 + \cdots + X_r$ where $X_1, \ldots, X_r \overset{\text{i.i.d.}}{\sim} \text{Geom}(p)$. Hence

$$\text{E}(X) = r \cdot \frac{q}{p}.$$

Proof. Let X_1 be the failures before the 1st success, X_2 the failures between the 1st and 2nd successes, and so on. Each block starts fresh, since the trials are independent, so the X_i are i.i.d. $\text{Geom}(p)$ and they add up to X. Linearity gives the mean. □

Example 4.17 (Coupon collector)

There are n types of toys, and each purchase gives a uniformly random type, independently. Let N be the number of purchases needed to get all n types. Find $\text{E}(N)$, and evaluate it for $n = 10$.

Solution. Break N into stages: $N = N_1 + N_2 + \cdots + N_n$, where N_j is the number of extra purchases needed to go from $j - 1$ types to j types. With $j - 1$ types already found, each purchase is new with probability

$$p_j = \frac{n-(j-1)}{n} = \frac{n-j+1}{n},$$

so $N_j \sim \text{FS}(p_j)$ and $\text{E}(N_j) = \frac{n}{n-j+1}$. By linearity,

$$\begin{aligned}
\text{E}(N) &= \sum_{j=1}^{n} \frac{n}{n-j+1} && \text{linearity} \\
&= n\left(\frac{1}{n} + \frac{1}{n-1} + \cdots + \frac{1}{1}\right) = n\sum_{k=1}^{n}\frac{1}{k} && \text{reverse the order} \\
&\approx n \ln n.
\end{aligned}$$

For $n = 10$: $\text{E}(N) = 10\left(1 + \frac{1}{2} + \cdots + \frac{1}{10}\right) \approx 29.3$ purchases. Most of that is the last few types: the final one alone takes 10 purchases on average.

Example 4.18 (St. Petersburg paradox)

Flip a fair coin until the first Heads. If that takes N flips, you win $X = 2^N$ dollars. Find $\text{E}(X)$. Then find it again if (a) you get nothing when the game goes past 40 flips, or (b) your winnings are capped at 2^{40}.

Solution. $N \sim \text{FS}(1/2)$, so $P(N = k) = (1/2)^k$, and

$$\text{E}(X) = \sum_{k=1}^{\infty} 2^k \left(\frac{1}{2}\right)^k = 1 + 1 + 1 + \cdots = \infty.$$

(a) If games past 40 flips pay 0,

$$\text{E}(X) = \sum_{k=1}^{40} 2^k \left(\frac{1}{2}\right)^k + \sum_{k=41}^{\infty} 0 = 40.$$

(b) If games past 40 flips pay 2^{40} instead,

$$\begin{aligned}
\text{E}(X) &= \sum_{k=1}^{40} 1 + 2^{40}\sum_{k=41}^{\infty}\left(\frac{1}{2}\right)^k = 40 + 2^{40}\cdot\left(\frac{1}{2}\right)^{40} && \text{geometric tail} \\
&= 41.
\end{aligned}$$

The infinite mean comes entirely from a tail of absurdly rare, absurdly huge payouts. Cut the tail anywhere realistic (and $2^{40} \approx 1.1 \times 10^{12}$ dollars is generous) and the value drops to around \$40.

Also note $\mathrm{E}(N) = 2$, so $2^{\mathrm{E}(N)} = 4$, while $\mathrm{E}(2^N) = \infty$. In general $\mathrm{E}(g(X)) \neq g(\mathrm{E}(X))$.

4.4 Indicator r.v.s and the Fundamental Bridge

Recall from Definition 3.13 that the indicator I_A is 1 if A happens and 0 otherwise.

Theorem 4.19 (Indicator Properties)

For events A and B:

1. $(I_A)^k = I_A$ for any positive integer k;
2. $I_{A^c} = 1 - I_A$;
3. $I_{A \cap B} = I_A \, I_B$;
4. $I_{A \cup B} = I_A + I_B - I_A \, I_B$.

Proof. Each one can be checked case by case, since indicators are only 0 or 1. For example, for (4): if neither happens both sides are 0; if exactly one happens both sides are 1; if both happen the right side is $1 + 1 - 1 = 1$. □

Theorem 4.20 (Fundamental Bridge)

For any event A,

$$P(A) = \mathrm{E}(I_A).$$

Proof. $I_A \sim \mathrm{Bern}(P(A))$, and a $\mathrm{Bern}(p)$ r.v. has mean p. □

The indicator method

To find $\mathrm{E}(X)$ when X *counts* something:

1. Write $X = I_1 + \cdots + I_n$, with one indicator for each thing that could be counted.
2. Find $\mathrm{E}(I_j) = P(\text{thing } j \text{ happens})$ by the fundamental bridge. Use symmetry if all j look the same.
3. Add up: $\mathrm{E}(X) = \sum_j P(\text{thing } j \text{ happens})$ by linearity.

The indicators can be as dependent as they like. You never need the distribution of X.

Example 4.21 (Boole's inequality and inclusion–exclusion)

Use indicators to show $P(A_1 \cup \cdots \cup A_n) \leq P(A_1) + \cdots + P(A_n)$, and prove inclusion–exclusion for two events.

Solution. $I_{A_1 \cup \cdots \cup A_n} \leq I_{A_1} + \cdots + I_{A_n}$: if the left side is 1, at least one term on the right is 1. Taking expectations and using monotonicity, linearity and the bridge,

$$P(A_1 \cup \cdots \cup A_n) \leq P(A_1) + \cdots + P(A_n).$$

For two events, take expectations of property (4) in Theorem 4.19:

$$P(A \cup B) = \mathrm{E}(I_A + I_B - I_A I_B) = P(A) + P(B) - P(A \cap B).$$

Example 4.22 (Card matching)

A deck of n cards labelled $1, \ldots, n$ is shuffled. Card j is a *match* if it ends up in position j. Find the expected number of matches.

Solution. Let $X = I_1 + \cdots + I_n$, where I_j indicates that card j is in position j. Card j is equally likely to be in any of the n positions, so $\mathrm{E}(I_j) = 1/n$. Then

$$\mathrm{E}(X) = \sum_{j=1}^{n} \frac{1}{n} = 1,$$

no matter how big n is. The I_j are dependent (if cards 1 to $n-1$ all match, card n must too), but

that doesn't matter for the mean.

Example 4.23 (Distinct birthdays and birthday matches)

In a group of n people with independent, uniformly random birthdays over 365 days, find (a) the expected number of distinct birthdays, and (b) the expected number of pairs of people who share a birthday.

Solution. (a) Put an indicator on each *day*: $X = \sum_{j=1}^{365} I_j$, where I_j indicates that at least one person was born on day j. Using the complement,

$$\mathrm{E}(I_j) = 1 - P(\text{nobody born on day } j) = 1 - \left(\frac{364}{365}\right)^n, \qquad \mathrm{E}(X) = 365\left(1 - \left(\frac{364}{365}\right)^n\right).$$

(b) Put an indicator on each *pair*: $Y = \sum_{i<j} J_{ij}$, where J_{ij} indicates that persons i and j match. There are $\binom{n}{2}$ pairs and each matches with probability $1/365$, so

$$\mathrm{E}(Y) = \frac{\binom{n}{2}}{365}.$$

For $n = 23$ this is $253/365 \approx 0.69$.

Example 4.24 (Local maxima of a permutation)

A random permutation $a_1, \ldots, a_n$ of $1, \ldots, n$ (with $n \geq 2$) has a *local maximum* at position j if a_j is bigger than its neighbours (just a_2 for $j = 1$, just a_{n-1} for $j = n$). Find the expected number of local maxima.

Solution. Let I_j indicate a local max at position j.

- Interior j ($2 \leq j \leq n-1$): a_j has to be the largest of the three values a_{j-1}, a_j, a_{j+1}. By symmetry each of the three is equally likely to be largest, so $\mathrm{E}(I_j) = 1/3$.
- Endpoints $j = 1, n$: a_j has to be the larger of two values, so $\mathrm{E}(I_j) = 1/2$.

By linearity,

$$\mathrm{E}\left(\sum_{j=1}^{n} I_j\right) = 2 \cdot \frac{1}{2} + (n-2) \cdot \frac{1}{3} = \frac{n+1}{3}.$$

Example 4.25 (Negative Hypergeometric)

An urn has w white and b black balls. We draw without replacement until we've drawn r white balls. Find the expected number of black balls drawn.

Solution. Label the black balls $1, \dots, b$ and let $X = I_1 + \cdots + I_b$, where I_j indicates that black ball j is drawn before the rth white ball.

Focus only on black ball j together with the w white balls, and ignore everything else. In a random order, black ball j is equally likely to land in any of the $w+1$ relative positions among these balls. It comes before the rth white ball exactly when it's in one of the first r positions, so

$$\mathrm{E}(I_j) = \frac{r}{w+1}, \qquad \mathrm{E}(X) = \frac{rb}{w+1}.$$

Sanity check: with $r = w$ (draw until all whites are out), the expected number of black balls left at the end is $b - \frac{wb}{w+1} = \frac{b}{w+1}$, which is the expected number of black balls after the last white one.

Theorem 4.26 (Expectation via the Survival Function)

If X takes values in $\{0, 1, 2, \dots\}$, then

$$\mathrm{E}(X) = \sum_{n=0}^{\infty} P(X > n).$$

Proof. Write X as a sum of indicators: $X = \sum_{k=1}^{\infty} I(X \geq k)$, since exactly X of these indicators equal 1. Then

$$\begin{aligned}
\mathrm{E}(X) &= \sum_{k=1}^{\infty} \mathrm{E}\big(I(X \geq k)\big) && \text{linearity} \\
&= \sum_{k=1}^{\infty} P(X \geq k) && \text{fundamental bridge} \\
&= \sum_{n=0}^{\infty} P(X > n). && n = k - 1
\end{aligned}$$

□

Example 4.27 (Geometric mean, the easy way)

For $X \sim \text{Geom}(p)$, Theorem 4.10 gives $P(X > n) = q^{n+1}$. So

$$\text{E}(X) = \sum_{n=0}^{\infty} q^{n+1} = \frac{q}{1-q} = \frac{q}{p},$$

with no derivatives needed.

4.5 Law of the Unconscious Statistician (LOTUS)

Theorem 4.28 (LOTUS)

If X is a discrete r.v. and $g : \mathbb{R} \to \mathbb{R}$, then

$$\text{E}\big(g(X)\big) = \sum_{x} g(x)\, P(X = x).$$

Proof. Write $\text{E}(g(X))$ as a sum over outcomes, then group the outcomes by the value of X:

$$\begin{aligned}
\text{E}\big(g(X)\big) &= \sum_{s \in S} g\big(X(s)\big)\, P(\{s\}) && \text{sum over outcomes} \\
&= \sum_{x} \sum_{s\,:\,X(s)=x} g\big(X(s)\big)\, P(\{s\}) && \text{group by value of } X \\
&= \sum_{x} g(x) \sum_{s\,:\,X(s)=x} P(\{s\}) && g(X(s)) = g(x) \text{ in each group} \\
&= \sum_{x} g(x)\, P(X = x).
\end{aligned}$$

□

Intuition. To get $\text{E}(g(X))$ we don't need the PMF of $g(X)$. Just keep the probabilities of X and swap each value x for $g(x)$. (It's "unconscious" because it's so natural people use it without realizing it needs proof.)

Example 4.29 (Second moment of a die)

For a fair die X, find $\mathrm{E}(X^2)$ and compare it to $(\mathrm{E}\,X)^2$.

Solution. By LOTUS with $g(x) = x^2$,

$$\mathrm{E}(X^2) = \sum_{j=1}^{6} j^2 \cdot \frac{1}{6} = \frac{1+4+9+16+25+36}{6} = \frac{91}{6} \approx 15.17,$$

but $(\mathrm{E}\,X)^2 = 3.5^2 = 12.25$. Not the same.

Pitfall. $\mathrm{E}(g(X)) \neq g(\mathrm{E}(X))$ in general. They're equal when g is linear, $g(x) = ax + b$, by linearity. For anything else (squares, exponentials, max, ...) you need LOTUS.

4.6 Variance

Definition 4.30 (Variance and Standard Deviation)

The **variance** of an r.v. X is

$$\mathrm{Var}(X) = \mathrm{E}\big[(X - \mathrm{E}\,X)^2\big],$$

and the **standard deviation** is $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$.

Intuition. Variance is the average squared distance from the mean, so it measures spread. We square so that deviations above and below the mean don't cancel. The SD undoes the squaring, so it's in the same units as X.

Theorem 4.31 (Shortcut Formula)

For any r.v. X,

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}\,X)^2.$$

Proof. Let $\mu = \mathrm{E}\,X$. Expanding the square,

$$\begin{aligned}\mathrm{E}\big[(X-\mu)^2\big] &= \mathrm{E}\big(X^2 - 2\mu X + \mu^2\big)\\ &= \mathrm{E}(X^2) - 2\mu\,\mathrm{E}(X) + \mu^2 && \text{linearity, } \mu \text{ is a constant}\\ &= \mathrm{E}(X^2) - 2\mu^2 + \mu^2 = \mathrm{E}(X^2) - \mu^2.\end{aligned}$$

□

Example 4.32 (Variance of a die)

For a fair die, Examples 4.2 and 4.29 give

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}\,X)^2 = \frac{91}{6} - \frac{49}{4} = \frac{182-147}{12} = \frac{35}{12} \approx 2.92,$$

so $\mathrm{SD}(X) \approx 1.71$.

Theorem 4.33 (Properties of Variance)

For any r.v. X and constant c:

1. $\mathrm{Var}(X+c) = \mathrm{Var}(X)$;
2. $\mathrm{Var}(cX) = c^2\,\mathrm{Var}(X)$;
3. if X and Y are **independent**, $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$;
4. $\mathrm{Var}(X) \geq 0$, with equality if and only if $P(X=a)=1$ for some constant a.

Proof. (1) Shifting X by c shifts the mean by c too, so $X + c - \mathrm{E}(X+c) = X - \mathrm{E}\,X$. (2) $\mathrm{Var}(cX) = \mathrm{E}\big[(cX - c\,\mathrm{E}\,X)^2\big] = c^2\,\mathrm{E}\big[(X-\mathrm{E}\,X)^2\big]$. (4) $\mathrm{Var}(X)$ is the mean of a nonnegative r.v., so by monotonicity it's ≥ 0, with equality only if $(X-\mathrm{E}\,X)^2 = 0$ with probability 1. (3) is proved in Chapter 7 using covariance. □

Pitfall. Variance is *not* linear. $\mathrm{Var}(cX) = c^2\,\mathrm{Var}(X)$, not $c\,\mathrm{Var}(X)$, and $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ needs independence (linearity of expectation doesn't). In particular $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ for independent X, Y, *not* the difference.

Example 4.34 (Binomial variance)

Let $X \sim \text{Bin}(n, p)$. Find $\text{Var}(X)$.

Solution. Write $X = I_1 + \cdots + I_n$ with I_j i.i.d. Bern(p). Since $I_j^2 = I_j$,

$$\text{Var}(I_j) = \text{E}(I_j^2) - (\text{E}\, I_j)^2 = p - p^2 = p(1-p).$$

The trials are independent, so the variances add:

$$\text{Var}(X) = \sum_{j=1}^{n} \text{Var}(I_j) = np(1-p) = npq.$$

Example 4.35 (Geometric and Negative Binomial variance)

Let $X \sim \text{Geom}(p)$. Find $\text{Var}(X)$, then use it to get the variance of $\text{NBin}(r, p)$.

Solution. From the proof of Theorem 4.11, $\sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1-q)^2}$. Multiply by q and differentiate again:

$$\sum_{k=1}^{\infty} kq^k = \frac{q}{(1-q)^2} \qquad \text{multiply by } q$$

$$\sum_{k=1}^{\infty} k^2 q^{k-1} = \frac{(1-q)^2 + 2q(1-q)}{(1-q)^4} = \frac{1+q}{(1-q)^3}. \qquad \text{quotient rule, cancel } (1-q)$$

By LOTUS,

$$\text{E}(X^2) = \sum_{k=0}^{\infty} k^2 q^k p = pq \cdot \frac{1+q}{p^3} = \frac{q(1+q)}{p^2},$$

so

$$\text{Var}(X) = \frac{q(1+q)}{p^2} - \frac{q^2}{p^2} = \frac{q}{p^2}.$$

Since $\text{NBin}(r, p)$ is a sum of r independent Geom(p) r.v.s (Theorem 4.16), the variances add:

$$\text{Var}(\text{NBin}(r, p)) = \frac{rq}{p^2}.$$

Also, $\text{FS}(p) = \text{Geom}(p) + 1$ has the same variance q/p^2, since shifting doesn't change variance.

4.7 Poisson

Definition 4.36 (Poisson Distribution)

An r.v. X has the **Poisson distribution** with parameter $\lambda > 0$, written $X \sim \text{Pois}(\lambda)$, if

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

This is a valid PMF because $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$.

Intuition. The Poisson is the go-to model for *counting rare events* when there are lots of chances for them to happen: typos on a page, emails in an hour, earthquakes in a year. λ is the average count.

Example 4.37 (Emails in an hour)

You get emails at an average rate of 3 per hour, modelled as $X \sim \text{Pois}(3)$. Find the probability of no emails in an hour, and of more than 3.

Solution.

$$P(X = 0) = e^{-3} \approx 0.050,$$

$$\begin{aligned} P(X > 3) &= 1 - e^{-3}\left(1 + 3 + \frac{3^2}{2!} + \frac{3^3}{3!}\right) && \text{complement} \\ &= 1 - 13\,e^{-3} \approx 0.353. \end{aligned}$$

Theorem 4.38 (Poisson Mean and Variance)

If $X \sim \text{Pois}(\lambda)$, then $\text{E}(X) = \lambda$ and $\text{Var}(X) = \lambda$.

Proof. **Mean.** The $k = 0$ term is 0, so

$$\mathrm{E}(X) = \sum_{k=1}^{\infty} k \frac{e^{-\lambda}\lambda^k}{k!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \qquad \text{cancel } k\text{, pull out } \lambda$$

$$= \lambda e^{-\lambda} e^{\lambda} = \lambda. \qquad \text{Taylor series of } e^{\lambda}$$

Variance. Differentiate $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$ with respect to λ, multiply by λ, and repeat:

$$\sum_{k=1}^{\infty} k \frac{\lambda^k}{k!} = \lambda e^{\lambda} \qquad \tfrac{d}{d\lambda}\text{, then } \times\lambda$$

$$\sum_{k=1}^{\infty} k^2 \frac{\lambda^k}{k!} = \lambda(1+\lambda)e^{\lambda}. \qquad \tfrac{d}{d\lambda}\text{, then } \times\lambda$$

So by LOTUS, $\mathrm{E}(X^2) = e^{-\lambda} \cdot \lambda(1+\lambda)e^{\lambda} = \lambda(1+\lambda)$, and

$$\mathrm{Var}(X) = \lambda(1+\lambda) - \lambda^2 = \lambda.$$

□

Poisson paradigm (approximation)

Let $A_1, \ldots, A_n$ be events with $p_j = P(A_j)$, where n is large, the p_j are small, and the A_j are independent or only weakly dependent. Then the number of A_j that happen,

$$X = \sum_{j=1}^{n} I(A_j),$$

is approximately Pois(λ) with

$$\lambda = \sum_{j=1}^{n} p_j = \mathrm{E}(X).$$

In particular, $P(\text{none of the } A_j \text{ happen}) = P(X = 0) \approx e^{-\lambda}$.

Intuition. This is what makes the Poisson so useful. You don't need to know the exact dependence structure. Find λ with indicators and linearity, which is easy, and then treat the count as Poisson.

Example 4.39 (Balls in boxes)

Put k distinguishable balls into n distinguishable boxes, each ball independently and uniformly. Find the expected number of empty boxes, and approximate the probability that at least one box is empty when $n = 1000$ and $k = 5806$.

Solution. Let I_j indicate that box j is empty. Each ball misses box j with probability $1 - \frac{1}{n}$, so

$$\mathrm{E}(I_j) = \left(1 - \frac{1}{n}\right)^k, \qquad \mathrm{E}(X) = n\left(1 - \frac{1}{n}\right)^k.$$

The exact probability that some box is empty comes from inclusion–exclusion,

$$P\left(\bigcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{n-1} (-1)^{j+1} \binom{n}{j} \left(1 - \frac{j}{n}\right)^k,$$

which is a pain to evaluate. Instead, with $n = 1000$ and $k = 5806$, $\mathrm{E}(X) = 1000(0.999)^{5806} \approx 3.00$. The empty-box events are only weakly dependent, so by the Poisson paradigm $X \approx \mathrm{Pois}(3)$:

$$P(X \geq 1) = 1 - P(X = 0) \approx 1 - e^{-3} \approx 0.95.$$

Example 4.40 (Birth-minute and birth-hour)

There are 1600 students. Assume birthdays and birth times are uniform and independent, with 365 days. Approximate the probability that (a) some two students were born in the same minute of the same day, and (b) some four students were born in the same hour of the same day.

Solution. (a) There are $c = 365 \cdot 24 \cdot 60 = 525{,}600$ day-minutes. Put an indicator on each of the $\binom{1600}{2}$ pairs; each pair matches with probability $1/c$. So

$$\lambda_1 = \binom{1600}{2} \frac{1}{525{,}600} \approx 2.434, \qquad P(\text{at least one pair}) \approx 1 - e^{-2.434} \approx 0.912.$$

(b) There are $b = 365 \cdot 24 = 8760$ day-hours.

Method 1 (indicators on groups of four). Four given people share a day-hour with probability $1/b^3$ (the first is anywhere, the other three must match). So

$$\lambda_2 = \binom{1600}{4} \frac{1}{8760^3} \approx 0.405, \qquad P \approx 1 - e^{-0.405} \approx 0.333.$$

Method 2 (indicators on day-hours). Let I_j indicate that day-hour j has at least 4 births. The number of births in a slot is $\text{Bin}(1600, 1/b)$, so with $p = 1/b$, $q = 1 - p$,

$$\text{E}(I_j) = 1 - \sum_{k=0}^{3} \binom{1600}{k} p^k q^{1600-k}, \qquad \lambda_3 = b\ \text{E}(I_1) \approx 0.350, \qquad P \approx 1 - e^{-0.350} \approx 0.295.$$

The exact answer is about 0.296, so Method 2 wins. Method 1's indicators are much more dependent: if people $1, 2, 3, 4$ share a day-hour, that makes it far more likely that $1, 2, 3, 5$ do too. Knowing one day-hour is crowded says almost nothing about another day-hour, so Method 2's indicators are close to independent, which is what the Poisson paradigm needs.

4.8 Connections between Poisson and Binomial

Theorem 4.41 (Sum of Independent Poissons)

If $X \sim \text{Pois}(\lambda_1)$ and $Y \sim \text{Pois}(\lambda_2)$ are independent, then $X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$.

Proof. Condition on X:

$$\begin{aligned}
P(X+Y=k) &= \sum_{j=0}^{k} P(X+Y=k \mid X=j)\, P(X=j) && \text{LOTP} \\
&= \sum_{j=0}^{k} P(Y=k-j)\, P(X=j) && \text{independence} \\
&= \sum_{j=0}^{k} \frac{e^{-\lambda_2}\lambda_2^{k-j}}{(k-j)!} \cdot \frac{e^{-\lambda_1}\lambda_1^{j}}{j!} && \text{Poisson PMFs} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{j=0}^{k} \binom{k}{j} \lambda_1^{j} \lambda_2^{k-j} && \text{multiply and divide by } k! \\
&= \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^k}{k!}. && \text{binomial theorem}
\end{aligned}$$

□

Theorem 4.42 (Poisson Given a Sum of Poissons)

If $X \sim \text{Pois}(\lambda_1)$ and $Y \sim \text{Pois}(\lambda_2)$ are independent, then

$$X \mid (X+Y=n) \;\sim\; \text{Bin}\left(n,\, \frac{\lambda_1}{\lambda_1+\lambda_2}\right).$$

Proof. By Bayes' rule and Theorem 4.41,

$$\begin{aligned} P(X=k \mid X+Y=n) &= \frac{P(Y=n-k)\,P(X=k)}{P(X+Y=n)} && \text{Bayes + independence} \\ &= \frac{\dfrac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}\cdot\dfrac{e^{-\lambda_1}\lambda_1^{k}}{k!}}{\dfrac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}} && \text{Poisson PMFs} \\ &= \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k\left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k}. && e\text{'s cancel} \end{aligned}$$

□

Intuition. Think of X and Y as counts of two types of event, like emails from work and emails from friends. Given n emails in total, each one is independently from work with probability proportional to its rate, $\frac{\lambda_1}{\lambda_1+\lambda_2}$. This mirrors Theorem 3.47, where conditioning two Binomials on their sum gave a Hypergeometric.

Example 4.43 (Splitting the emails)

Work emails arrive as Pois(2) per hour and personal ones as Pois(1) per hour, independently. Given 6 emails in an hour, what's the probability exactly 4 were work emails?

Solution. By Theorem 4.42, given a total of 6, the number of work emails is $\text{Bin}(6, 2/3)$:

$$P(4 \text{ work} \mid 6 \text{ total}) = \binom{6}{4}\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)^2 = 15\cdot\frac{16}{729} = \frac{240}{729} \approx 0.329.$$

Theorem 4.44 (Poisson Approximation to Binomial)

If $X \sim \text{Bin}(n, p)$ and $n \to \infty$, $p \to 0$ with $np = \lambda$ fixed, then for each $k \geq 0$,

$$P(X = k) \to \frac{e^{-\lambda}\lambda^k}{k!}.$$

Proof. Substitute $p = \lambda/n$ and regroup:

$$P(X = k) = \binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1 - \frac{\lambda}{n}\right)^{n-k}$$

$$= \frac{\lambda^k}{k!} \cdot \frac{n(n-1)\cdots(n-k+1)}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-k}. \quad \text{split } \tbinom{n}{k} \text{ and the power}$$

As $n \to \infty$ with k fixed:

$$\frac{n(n-1)\cdots(n-k+1)}{n^k} \to 1, \qquad \left(1 - \frac{\lambda}{n}\right)^n \to e^{-\lambda}, \qquad \left(1 - \frac{\lambda}{n}\right)^{-k} \to 1.$$

Multiplying the limits gives $\frac{e^{-\lambda}\lambda^k}{k!}$. □

Example 4.45 (How good is the approximation?)

Compare $P(X = 0)$ for $X \sim \text{Bin}(n, 2/n)$ against Pois(2) as n grows.

Solution. $P(X = 0) = (1 - 2/n)^n$, while the Poisson value is $e^{-2} \approx 0.1353$:

n	10	50	1000	Pois(2)
$P(X = 0)$	0.1074	0.1299	0.1351	0.1353

By $n = 1000$ ($p = 0.002$) the two are basically identical.

Example 4.46 (Visitors to a website)

A website has 10^6 potential visitors per day, each independently visiting with probability 2×10^{-6}. Approximate the probability of at least 3 visitors in a day.

Solution. The number of visitors is $\text{Bin}(10^6, 2 \times 10^{-6})$, which is approximately $\text{Pois}(2)$ since $\lambda = np = 2$. Then

$$\begin{aligned} P(X \geq 3) &= 1 - P(X = 0) - P(X = 1) - P(X = 2) && \text{complement} \\ &\approx 1 - e^{-2}\left(1 + 2 + \frac{2^2}{2!}\right) && \text{Poisson PMF} \\ &= 1 - 5e^{-2} \approx 0.3233. \end{aligned}$$

4.9 *Using Probability and Expectation to Prove Existence

The **probabilistic method**: to show that an object with some property exists, pick an object *at random* and show that it has the property with positive probability. We never have to construct the object.

The two principles

- **Possibility principle.** If a randomly chosen object has the property with probability $P(A) > 0$, then some object has the property.
- **Good score principle.** If a randomly chosen object has score X with $\text{E}(X) \geq c$, then some object has score at least c. (Not everything can be below average.) Similarly, some object has score $\leq \text{E}(X)$.

If scores are integers and $\text{E}(X) > c$, some object has score at least $\lfloor c \rfloor + 1$.

Example 4.47 (Overlapping committees)

100 people are split into 15 committees of 20, so that each person is on exactly 3 committees. Show that some two committees share at least 3 members.

Solution. Pick two of the 15 committees at random and let X be the number of people on both. Write $X = I_1 + \cdots + I_{100}$, where I_j indicates that person j is on both chosen committees.

Person j is on 3 of the 15 committees. Picking 2 committees is like recapturing 2 elk from 15, with 3 tagged, so by the Hypergeometric PMF,

$$\mathrm{E}(I_j) = \frac{\binom{3}{2}\binom{12}{0}}{\binom{15}{2}} = \frac{3}{105} = \frac{1}{35}, \qquad \mathrm{E}(X) = \frac{100}{35} = \frac{20}{7} \approx 2.86.$$

By the good score principle, some pair of committees has overlap at least $20/7$. Overlaps are integers, so that pair shares at least 3 people.

Example 4.48 (LYM inequality)

Let $\mathcal{F}$ be a family of subsets of $\{1, \ldots, n\}$ in which no set contains another, and let s_k be the number of sets in $\mathcal{F}$ of size k. Show that

$$\sum_{k=0}^{n} \frac{s_k}{\binom{n}{k}} \leq 1.$$

Solution. Take a uniformly random permutation of $1, \ldots, n$, and look at its *prefixes*: the sets formed by its first k elements, for $k = 0, \ldots, n$. For each $F \in \mathcal{F}$, let A_F be the event that F is one of these prefixes.

If $|F| = k$, then by symmetry the first k elements are a uniformly random k-subset, so

$$P(A_F) = \frac{1}{\binom{n}{k}}.$$

The prefixes are nested (each one contains the previous), and no set in $\mathcal{F}$ contains another, so *at most one* set of $\mathcal{F}$ can be a prefix. That means the A_F are disjoint, and

$$\sum_{F \in \mathcal{F}} P(A_F) = P\left(\bigcup_{F \in \mathcal{F}} A_F\right) \leq 1.$$

Grouping the sets by size, the left side is $\sum_k s_k / \binom{n}{k}$.

Theorem 4.49 (Ramsey Lower Bound)

Let $R(k, k)$ be the smallest n such that every red/blue colouring of the edges of the complete graph K_n contains a K_k with all edges the same colour. If

$$\binom{n}{k} 2^{1-\binom{k}{2}} < 1,$$

then $R(k, k) > n$. In particular, $R(k, k) > 2^{k/2}$ for $k \geq 3$.

Proof. Colour each edge of K_n red or blue independently with probability $1/2$. For each set S of k vertices, let I_S indicate that all $\binom{k}{2}$ edges inside S have the same colour:

$$\mathrm{E}(I_S) = 2\left(\frac{1}{2}\right)^{\binom{k}{2}} = 2^{1-\binom{k}{2}}.$$

Let $X = \sum_{|S|=k} I_S$ count the single-colour K_k's. By linearity, $\mathrm{E}(X) = \binom{n}{k} 2^{1-\binom{k}{2}} < 1$. Since X is a nonnegative integer with mean below 1, $P(X = 0) > 0$. So some colouring has no single-colour K_k, which means $R(k, k) > n$.

For the "in particular", take $n = \lfloor 2^{k/2} \rfloor$. Using $\binom{n}{k} \le \frac{n^k}{k!} \le \frac{2^{k^2/2}}{k!}$,

$$\binom{n}{k} 2^{1-\binom{k}{2}} \le \frac{2^{k^2/2}}{k!} \cdot 2^{1-\frac{k^2}{2}+\frac{k}{2}} = \frac{2^{1+k/2}}{k!},$$

which is less than 1 for all $k \ge 3$ (at $k = 3$ it's $2^{2.5}/6 \approx 0.94$, and the $k!$ grows much faster than $2^{k/2}$). □

4.10 Recap

Chapter 4 at a glance

- $\mathrm{E}(X) = \sum_x x\, P(X = x)$ is a weighted average. It only depends on the distribution.
- **Linearity:** $\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$ always holds, even for dependent r.v.s.
- **Fundamental bridge:** $P(A) = \mathrm{E}(I_A)$. Combined with linearity, it turns "expected number of things" into "sum of probabilities".
- **LOTUS:** $\mathrm{E}(g(X)) = \sum_x g(x)\, P(X = x)$. No need for the distribution of $g(X)$.
- **Variance:** $\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}\, X)^2$. Not linear: $\mathrm{Var}(cX) = c^2 \mathrm{Var}(X)$, and variances only add for independent r.v.s.
- **Poisson paradigm:** Counts of many rare, weakly dependent events are approximately $\mathrm{Pois}(\lambda)$ with $\lambda =$ expected count.

Discrete distributions: means and variances

Distribution	Story	Mean	Variance
Bern(p)	one trial	p	pq
Bin(n, p)	# successes in n trials	np	npq
HGeom(w, b, n)	# white, no replacement	$\frac{nw}{w+b}$	(Chapter 7)
Geom(p)	failures before 1st success	$\frac{q}{p}$	$\frac{q}{p^2}$
FS(p)	trials until 1st success	$\frac{1}{p}$	$\frac{q}{p^2}$
NBin(r, p)	failures before rth success	$\frac{rq}{p}$	$\frac{rq}{p^2}$
Pois(λ)	count of rare events	λ	λ

Common mistakes

1. Writing $\mathrm{E}(g(X)) = g(\mathrm{E}\,X)$ for non-linear g (see St. Petersburg).
2. Adding variances of dependent r.v.s, or writing $\mathrm{Var}(cX) = c\,\mathrm{Var}(X)$.
3. Thinking linearity needs independence. It doesn't.
4. Mixing up Geom (counts failures, starts at 0) and FS (counts trials, starts at 1).
5. Using $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y)$ without independence.

Chapter 5: Continuous Random Variables

5.1 Probability Density Functions

A discrete r.v. takes values you can list. A **continuous** r.v. can take any value in an interval, like $(0, 1)$, $(0, \infty)$ or all of $\mathbb{R}$. Its CDF has no jumps, so instead of a PMF we describe it with a *density*.

Definition 5.1 (Continuous r.v.)

An r.v. has a **continuous distribution** if its CDF is differentiable. We also allow finitely many points (like endpoints) where the CDF is continuous but not differentiable, as long as it's differentiable everywhere else. A **continuous r.v.** is an r.v. with a continuous distribution.

Definition 5.2 (Probability Density Function)

For a continuous r.v. X with CDF F, the **probability density function** (PDF) of X is the derivative of the CDF:

$$f(x) = F'(x).$$

The **support** of X is the set of all x where $f(x) > 0$.

Remark. For a continuous r.v., $P(X = x) = 0$ for *every* x. The probability of a single value is the height of the jump in the CDF at that value, and a continuous CDF has no jumps. That's why a PMF would be useless here (it would be 0 everywhere) and we use a PDF instead.

Intuition. $f(x)$ is *not* a probability, and it can be bigger than 1. It's a probability *per unit length*. For a tiny interval of length ε around x,

$$P\big(x - \tfrac{\varepsilon}{2} < X < x + \tfrac{\varepsilon}{2}\big) = \int_{x-\varepsilon/2}^{x+\varepsilon/2} f(t)\,dt \approx f(x)\,\varepsilon,$$

since f is roughly constant on a tiny interval. So think of $f(x)\,dx$ as "the probability that X lands in a tiny interval of length dx at x". Units check: if X is in cm, then $f(x)$ is in cm^{-1}, and multiplying by dx (in cm) gives a unitless probability.

Proposition 5.3 (PDF to CDF)

Let X be a continuous r.v. with PDF f. Then its CDF is

$$F(x) = \int_{-\infty}^{x} f(t)\,dt.$$

Proof. By definition, F is an antiderivative of f, so by the fundamental theorem of calculus,

$$\int_{-\infty}^{x} f(t)\,dt = F(x) - \lim_{t\to-\infty} F(t) \qquad \text{FTC}$$

$$= F(x) - 0 = F(x). \qquad \text{CDF} \to 0 \text{ at } -\infty$$

□

Intuition. This mirrors the discrete case, where $F(x)$ was the sum of the PMF over all values $\leq x$. Here $F(x)$ is the *area* under the PDF to the left of x. Differentiating and integrating undo each other, so the PDF and CDF carry exactly the same information.

Getting probabilities from a PDF

To get a probability, **integrate the PDF over the region you care about**:

$$P(a < X \leq b) = F(b) - F(a) = \int_{a}^{b} f(x)\,dx, \qquad \text{interval}$$

$$P(X \in A) = \int_{A} f(x)\,dx. \qquad \text{any region } A \subseteq \mathbb{R}$$

Since single points have probability 0, endpoints don't matter:

$$P(a < X < b) = P(a < X \leq b) = P(a \leq X < b) = P(a \leq X \leq b).$$

Pitfall. Being carefree about endpoints is only OK for *continuous* r.v.s. For a discrete r.v., $P(X \leq 3)$ and $P(X < 3)$ can be different, since $P(X = 3)$ can be positive.

Theorem 5.4 (Valid PDFs)

The PDF f of a continuous r.v. must satisfy:

1. **Nonnegative:** $f(x) \geq 0$ for all x;
2. **Integrates to 1:** $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$.

Conversely, any function with these two properties is the PDF of some continuous r.v.

Proof. (1) $f(x)$ is the slope of the CDF at x, and a CDF can never decrease, so its slope can't be negative. (Or: if $f(x_0) < 0$, integrating over a tiny interval around x_0 would give a negative probability.) (2) The integral over all of $\mathbb{R}$ is $P(-\infty < X < \infty) = 1$. □

Example 5.5 (Finding the constant, and a density above 1)

Let $f(x) = c\,x^2$ for $0 \leq x \leq 2$, and $f(x) = 0$ otherwise. Find c so that f is a valid PDF, then find the CDF and $P(X > 1)$.

Solution. For $c > 0$, f is nonnegative, so we just need the total area to be 1:

$$1 = \int_0^2 c\,x^2\,dx \qquad \text{integrates to 1}$$

$$= c \cdot \frac{x^3}{3}\bigg|_0^2 = \frac{8c}{3},$$

so $c = \frac{3}{8}$. Notice $f(2) = \frac{3}{8} \cdot 4 = \frac{3}{2} > 1$, which is fine for a density.

For $0 \leq x \leq 2$, the CDF is

$$F(x) = \int_0^x \frac{3}{8}t^2\,dt = \frac{x^3}{8},$$

with $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 2$. Then

$$P(X > 1) = 1 - F(1) = 1 - \frac{1}{8} = \frac{7}{8}.$$

Example 5.6 (Logistic distribution)

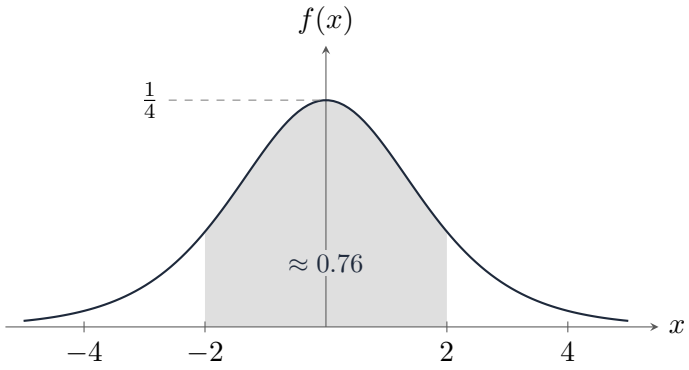
The **Logistic** distribution has CDF

$$F(x) = \frac{e^x}{1+e^x}, \qquad x \in \mathbb{R}.$$

Find the PDF, and find $P(-2 < X < 2)$ for $X \sim$ Logistic.

Solution. Differentiate the CDF with the quotient rule:

$$f(x) = \frac{e^x(1+e^x) - e^x \cdot e^x}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2}, \qquad x \in \mathbb{R}.$$



The shaded area is $P(-2 < X < 2)$. Since we already have F as an antiderivative,

$$P(-2 < X < 2) = F(2) - F(-2) = \frac{e^2}{1+e^2} - \frac{e^{-2}}{1+e^{-2}} \approx 0.881 - 0.119 = 0.76.$$

Without knowing F. Substitute $u = 1 + e^x$, so $du = e^x\,dx$, and the limits $x = \pm 2$ become $u = 1 + e^{\pm 2}$:

$$\int_{-2}^{2} \frac{e^x}{(1+e^x)^2}\,dx = \int_{1+e^{-2}}^{1+e^2} \frac{1}{u^2}\,du = \left[-\frac{1}{u}\right]_{1+e^{-2}}^{1+e^2} \approx 0.76.$$

Example 5.7 (Rayleigh distribution)

The **Rayleigh** distribution has CDF

$$F(x) = 1 - e^{-x^2/2}, \qquad x > 0.$$

Find the PDF and $P(X > 2)$.

Solution. By the chain rule,

$$f(x) = F'(x) = -e^{-x^2/2} \cdot (-x) = x\, e^{-x^2/2}, \qquad x > 0.$$

We could integrate f from 2 to ∞ with $u = x^2/2$, but F already does the work:

$$P(X > 2) = \int_2^\infty x\, e^{-x^2/2}\, dx = 1 - F(2) = e^{-2} \approx 0.135.$$

Definition 5.8 (Expectation of a Continuous r.v.)

The **expected value** of a continuous r.v. X with PDF f is

$$\mathrm{E}(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx,$$

as long as $\int_{-\infty}^{\infty} |x|\, f(x)\, dx < \infty$. In practice we only integrate over the support, since $f = 0$ elsewhere.

Intuition. Same idea as the discrete case: replace the sum with an integral and the PMF with the PDF. $\mathrm{E}(X)$ is still the balancing point of the distribution. The units work out too: $x\, f(x)\, dx$ has units $\mathrm{cm} \cdot \mathrm{cm}^{-1} \cdot \mathrm{cm} = \mathrm{cm}$.

Example 5.9 (Mean of the x^2 density)

Find $\mathrm{E}(X)$ for the PDF $f(x) = \frac{3}{8}x^2$ on $[0, 2]$ from Example 5.5.

Solution.

$$\mathrm{E}(X) = \int_0^2 x \cdot \frac{3}{8}x^2\, dx = \frac{3}{8} \cdot \frac{x^4}{4}\bigg|_0^2 = \frac{3}{8} \cdot 4 = \frac{3}{2}.$$

The mean is above the midpoint 1, which makes sense since the density piles up toward 2.

Example 5.10 (Mean of the Logistic, by symmetry)

Show that $X \sim$ Logistic has $\mathrm{E}(X) = 0$.

Solution. The PDF is symmetric about 0: multiplying top and bottom by e^{2x},

$$f(-x) = \frac{e^{-x}}{(1+e^{-x})^2} = \frac{e^{x}}{(e^{x}+1)^2} = f(x).$$

The tails of f shrink like $e^{-|x|}$, so $\int |x|\, f(x)\, dx < \infty$ and the mean exists. For a symmetric density, the contributions $x\, f(x)$ and $(-x)\, f(-x)$ cancel, so $\mathrm{E}(X) = 0$.

Theorem 5.11 (LOTUS, Continuous)

If X is a continuous r.v. with PDF f and $g : \mathbb{R} \to \mathbb{R}$, then

$$\mathrm{E}\big(g(X)\big) = \int_{-\infty}^{\infty} g(x)\, f(x)\, dx.$$

Proof. We prove the case where g is strictly increasing and differentiable. Let $Y = g(X)$, and let (a, b) be the range of g (possibly infinite). Since g is strictly increasing, g^{-1} exists and is strictly increasing too, so for $y \in (a, b)$,

$$F_Y(y) = P(g(X) \le y) = P\big(X \le g^{-1}(y)\big) = F_X\big(g^{-1}(y)\big). \qquad g \text{ increasing}$$

Differentiating with the chain rule,

$$f_Y(y) = f_X\big(g^{-1}(y)\big) \cdot \frac{d}{dy}\, g^{-1}(y), \qquad a < y < b,$$

and $f_Y(y) = 0$ outside (a, b). So

$$\mathrm{E}(Y) = \int_a^b y\, f_X\big(g^{-1}(y)\big) \frac{d}{dy}\, g^{-1}(y)\, dy.$$

Now substitute $x = g^{-1}(y)$. Then $y = g(x)$, $dx = \frac{d}{dy} g^{-1}(y)\, dy$, and as y runs from a to b, x runs from $-\infty$ to ∞:

$$\mathrm{E}\big(g(X)\big) = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx.$$

The general case (any g) holds too, but needs more machinery. □

Remark. Variance is defined exactly as before, $\mathrm{Var}(X) = \mathrm{E}\big[(X - \mathrm{E}\,X)^2\big]$, and the shortcut $\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}\,X)^2$ still holds. With continuous LOTUS, $\mathrm{E}(X^2) = \int x^2 f(x)\, dx$.

Example 5.12 (Variance of the x^2 density)

Find $\mathrm{Var}(X)$ for $f(x) = \frac{3}{8}x^2$ on $[0, 2]$.

Solution. By continuous LOTUS with $g(x) = x^2$,

$$\mathrm{E}(X^2) = \int_0^2 x^2 \cdot \frac{3}{8}x^2 \, dx = \frac{3}{8} \cdot \frac{x^5}{5}\bigg|_0^2 = \frac{3}{8} \cdot \frac{32}{5} = \frac{12}{5}.$$

Using $\mathrm{E}(X) = \frac{3}{2}$ from Example 5.9,

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}\,X)^2 = \frac{12}{5} - \frac{9}{4} = \frac{48 - 45}{20} = \frac{3}{20}.$$

5.2 Uniform

A uniform r.v. on the interval (a, b) is a completely random number between a and b.

Definition 5.13 (Uniform Distribution)

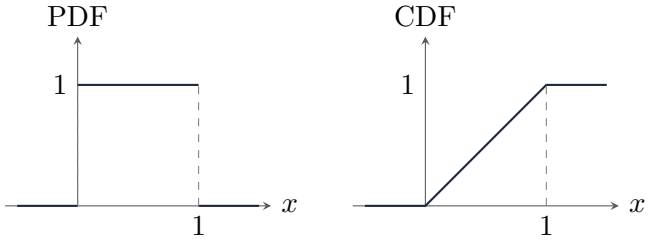
A continuous r.v. U has the **Uniform distribution** on (a, b), written $U \sim \mathrm{Unif}(a, b)$, if its PDF is

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise.} \end{cases}$$

The CDF is the accumulated area under the PDF:

$$F(x) = \begin{cases} 0, & x \le a, \\ \dfrac{x-a}{b-a}, & a < x < b, \\ 1, & x \ge b. \end{cases}$$

$\mathrm{Unif}(0, 1)$ is called the **standard Uniform** distribution.



$\mathrm{Unif}(0, 1)$ PDF (left) and CDF (right).

Remark. For $U \sim \text{Unif}(a, b)$ and any subinterval (c, d) of (a, b),

$$P\big(U \in (c, d)\big) = \frac{d - c}{b - a},$$

so probability is *proportional to length*. For $\text{Unif}(0, 1)$, probability *is* length.

Proposition 5.14 (Conditioning a Uniform)

Let $U \sim \text{Unif}(a, b)$ and let (c, d) be a subinterval of (a, b). Then the conditional distribution of U given $U \in (c, d)$ is $\text{Unif}(c, d)$.

Proof. For $c < u < d$,

$$\begin{aligned} P\big(U \le u \mid U \in (c, d)\big) &= \frac{P\big(U \le u,\ c < U < d\big)}{P\big(U \in (c, d)\big)} && \text{def. of cond. prob.} \\ &= \frac{P\big(U \in (c, u]\big)}{P\big(U \in (c, d)\big)} && \text{both events at once} \\ &= \frac{(u - c)/(b - a)}{(d - c)/(b - a)} = \frac{u - c}{d - c}, && \text{prob. } \propto \text{ length} \end{aligned}$$

which is the $\text{Unif}(c, d)$ CDF. □

Example 5.15 (Zooming in on a Uniform)

Let $U \sim \text{Unif}(0, 1)$. Find $P(0.2 < U < 0.6)$, and then $P(U < 0.5 \mid 0.2 < U < 0.6)$.

Solution. Probability is length, so $P(0.2 < U < 0.6) = 0.4$. Given $U \in (0.2, 0.6)$, $U \sim \text{Unif}(0.2, 0.6)$ by Proposition 5.14, so

$$P(U < 0.5 \mid 0.2 < U < 0.6) = \frac{0.5 - 0.2}{0.6 - 0.2} = \frac{0.3}{0.4} = \frac{3}{4}.$$

Theorem 5.16 (Uniform Mean and Variance)

If $U \sim \text{Unif}(a, b)$, then

$$\text{E}(U) = \frac{a + b}{2}, \qquad \text{Var}(U) = \frac{(b - a)^2}{12}.$$

Proof. The mean is the midpoint, as expected since the PDF is flat:

$$\text{E}(U) = \int_a^b x \cdot \frac{1}{b - a}\, dx = \frac{1}{b - a}\left(\frac{b^2}{2} - \frac{a^2}{2}\right) = \frac{a + b}{2}.$$

For the variance, first find $\mathrm{E}(U^2)$ with continuous LOTUS:

$$\mathrm{E}(U^2) = \int_a^b x^2 \cdot \frac{1}{b-a}\,dx = \frac{1}{3}\cdot\frac{b^3-a^3}{b-a} \qquad \text{LOTUS}$$

$$= \frac{a^2+ab+b^2}{3}. \qquad b^3-a^3 = (b-a)(a^2+ab+b^2)$$

Then

$$\mathrm{Var}(U) = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12} \qquad \text{common denominator}$$

$$= \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12}.$$

□

Definition 5.17 (Location-Scale Transformation)

Let X be an r.v. and $Y = \sigma X + \mu$, where σ and μ are constants with $\sigma > 0$. We say Y is obtained from X by a **location-scale transformation**: μ controls how the location is changed (shift) and σ controls how the scale is changed (stretch).

Example 5.18 (Uniform mean and variance, the easy way)

Use a location-scale transformation to get Theorem 5.16 without the algebra.

Solution. Start with $U \sim \mathrm{Unif}(0,1)$, whose PDF is 1 on $(0,1)$:

$$\mathrm{E}(U) = \int_0^1 x\,dx = \frac{1}{2}, \qquad \mathrm{E}(U^2) = \int_0^1 x^2\,dx = \frac{1}{3}, \qquad \mathrm{Var}(U) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

To turn U into a general $\mathrm{Unif}(a,b)$ r.v., stretch the support from length 1 to length $b-a$ (multiply by $b-a$, giving $\mathrm{Unif}(0, b-a)$), then shift it to start at a:

$$\tilde{U} = a + (b-a)U \sim \mathrm{Unif}(a,b).$$

Now linearity and the properties of variance do the work:

$$\mathrm{E}(\tilde{U}) = a + (b-a)\,\mathrm{E}(U) = a + \frac{b-a}{2} = \frac{a+b}{2}, \qquad \text{linearity}$$

$$\mathrm{Var}(\tilde{U}) = (b-a)^2\,\mathrm{Var}(U) = \frac{(b-a)^2}{12}. \qquad \text{shift drops out, scale comes out squared}$$

5.3 Universality of the Uniform

Given a $\mathrm{Unif}(0,1)$ r.v., we can construct an r.v. with any continuous distribution we want. Conversely, given an r.v. with any continuous distribution, we can create a $\mathrm{Unif}(0,1)$ r.v. We call this the

universality of the Uniform, because it tells us the Uniform is a universal starting point for building r.v.s with other distributions. (It's also called the probability integral transform, inverse transform sampling, or the quantile transformation.)

Theorem 5.19 (Universality of the Uniform)

Let F be a CDF that is continuous and strictly increasing on the support, so F^{-1} exists, and let $U \sim \text{Unif}(0,1)$.

1. If $X = F^{-1}(U)$, then X has CDF F.
2. If X has CDF F, then $F(X) \sim \text{Unif}(0,1)$.

Proof. (1) For any x,

$$P(X \leq x) = P\big(F^{-1}(U) \leq x\big) = P\big(U \leq F(x)\big) \qquad \text{apply } F \text{ to both sides; } F \text{ increasing}$$

$$= F(x). \qquad P(U \leq u) = u \text{ for } u \in (0,1)$$

(2) For $0 < u < 1$,

$$P\big(F(X) \leq u\big) = P\big(X \leq F^{-1}(u)\big) = F\big(F^{-1}(u)\big) = u,$$

which is the $\text{Unif}(0,1)$ CDF. □

Intuition. Recipe for part (1): take the CDF you're interested in, take its inverse, and plug in the Uniform. U picks a random height between 0 and 1 on the CDF's vertical axis, and F^{-1} reads off which x is at that height. Steep parts of F (where the PDF is big) cover lots of heights over a short stretch of x, so X lands there more often.

Example 5.20 (Simulating a Logistic)

Using one $U \sim \text{Unif}(0,1)$, construct an r.v. with the Logistic distribution from Example 5.6.

Solution. Invert $F(x) = \frac{e^x}{1+e^x}$ by solving $u = \frac{e^x}{1+e^x}$ for x:

$$u(1+e^x) = e^x \implies e^x = \frac{u}{1-u} \implies F^{-1}(u) = \ln\left(\frac{u}{1-u}\right).$$

So by Theorem 5.19, $X = \ln\left(\frac{U}{1-U}\right) \sim$ Logistic.

5.4 Normal

The Normal is a continuous distribution with a bell-shaped PDF. Once we have the *standard* Normal, we'll be able to get any Normal distribution we want by shifting and scaling.

Definition 5.21 (Standard Normal Distribution)

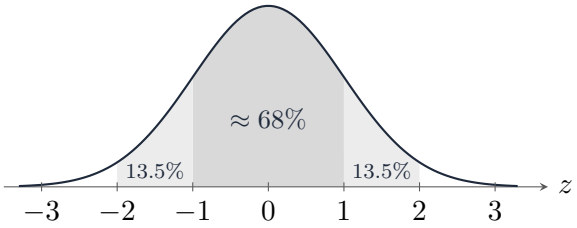
A continuous r.v. Z has the **standard Normal distribution**, written $Z \sim \mathcal{N}(0,1)$, if its PDF is

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad -\infty < z < \infty.$$

The standard Normal CDF Φ is the accumulated area under the PDF:

$$\Phi(z) = \int_{-\infty}^{z} \varphi(t)\, dt = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\, dt.$$

Remark. There is no closed-form expression for the antiderivative of φ: we can't write Φ in terms of elementary functions. So Φ is just *defined* as this integral, and in practice we get its values from software or a table.



Standard Normal PDF φ. Dark: within 1 SD of the mean; dark + light: within 2 SDs ($\approx 95\%$).

Symmetry properties of the standard Normal

1. **Symmetry of the PDF:** $\varphi(z) = \varphi(-z)$.
2. **Symmetry of tail areas:** the area to the left of $-z$ equals the area to the right of z, i.e. $P(Z \le -z) = P(Z \ge z)$, so
$$\Phi(z) = 1 - \Phi(-z) \quad \text{for all } z.$$
3. **Symmetry of Z and $-Z$:** if $Z \sim \mathcal{N}(0,1)$, then $-Z \sim \mathcal{N}(0,1)$ as well.

Proof. (1) z only appears as z^2. (2) Substitute $u = -t$ and use that the PDF integrates to 1:

$$\Phi(-z) = \int_{-\infty}^{-z} \varphi(t)\, dt = \int_{z}^{\infty} \varphi(u)\, du = 1 - \int_{-\infty}^{z} \varphi(u)\, du = 1 - \Phi(z).$$

(3) $P(-Z \le z) = P(Z \ge -z) = 1 - \Phi(-z) = \Phi(z)$, using (2). So $-Z$ has CDF Φ. □

Theorem 5.22 (φ is a Valid PDF)
$\int_{-\infty}^{\infty} e^{-z^2/2}\,dz = \sqrt{2\pi}$, so φ integrates to 1.

Proof. We can't integrate $e^{-z^2/2}$ directly, so we square the integral and switch to polar coordinates. Since z is a dummy variable, call it x in one copy and y in the other:

$$\begin{aligned}
\left(\int_{-\infty}^{\infty} e^{-z^2/2}\,dz\right)^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2/2}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}\,dy\right) && \text{rename dummy variables}\\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\,dx\,dy && \text{combine into a double integral}\\
&= \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2}\,r\,dr\,d\theta && \text{polar; extra } r \text{ is the Jacobian}\\
&= \int_0^{2\pi}\left(\int_0^{\infty} e^{-u}\,du\right)d\theta && u = r^2/2,\ du = r\,dr\\
&= \int_0^{2\pi} 1\,d\theta = 2\pi.
\end{aligned}$$

Taking square roots, the integral is $\sqrt{2\pi}$. □

Theorem 5.23 (Standard Normal Mean and Variance)
If $Z \sim \mathcal{N}(0,1)$, then $\mathrm{E}(Z) = 0$ and $\mathrm{Var}(Z) = 1$.

Proof. **Mean.** $g(z) = z\,e^{-z^2/2}$ is an odd function ($g(-z) = -g(z)$), and the integral of an odd function over a symmetric range is 0:

$$\mathrm{E}(Z) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} z\,e^{-z^2/2}\,dz = 0.$$

Variance. Since $\mathrm{E}(Z) = 0$, $\mathrm{Var}(Z) = \mathrm{E}(Z^2)$. The integrand $z^2 e^{-z^2/2}$ is even, so

$$\begin{aligned}
\mathrm{Var}(Z) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz = \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} z^2 e^{-z^2/2}\, dz && \text{LOTUS; even function} \\
&= \frac{2}{\sqrt{2\pi}} \left(\left[-z\, e^{-z^2/2} \right]_0^{\infty} + \int_0^{\infty} e^{-z^2/2}\, dz \right) && \text{parts: } u = z,\ dv = z e^{-z^2/2} dz \\
&= \frac{2}{\sqrt{2\pi}} \left(0 + \frac{\sqrt{2\pi}}{2} \right) = 1. && \text{Thm 5.22, half the area}
\end{aligned}$$

□

Definition 5.24 (Normal Distribution)

If $Z \sim \mathcal{N}(0,1)$, then

$$X = \mu + \sigma Z$$

has the **Normal distribution** with mean μ and variance σ^2, for any real μ and $\sigma > 0$. We write $X \sim \mathcal{N}(\mu, \sigma^2)$.

Remark. This is a location-scale transformation of Z, so by linearity and the variance properties, $\mathrm{E}(X) = \mu + \sigma \cdot 0 = \mu$ and $\mathrm{Var}(X) = \sigma^2 \cdot 1 = \sigma^2$. Going the other way, **standardization**: if $X \sim \mathcal{N}(\mu, \sigma^2)$ then

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0,1).$$

Theorem 5.25 (Normal CDF and PDF)

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then the CDF and PDF of X are

$$F(x) = \Phi\left(\frac{x-\mu}{\sigma} \right), \qquad f(x) = \varphi\left(\frac{x-\mu}{\sigma} \right) \frac{1}{\sigma} = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} \right).$$

Proof. Standardize inside the probability:

$$\begin{aligned}
F(x) = P(X \le x) &= P\left(\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma} \right) && \text{same event, } \sigma > 0 \\
&= \Phi\left(\frac{x-\mu}{\sigma} \right). && \text{standardized } X \text{ is } \mathcal{N}(0,1)
\end{aligned}$$

Differentiating with the chain rule gives $f(x) = \varphi\big(\frac{x-\mu}{\sigma}\big) \cdot \frac{1}{\sigma}$, and plugging in the formula for φ gives the last form. □

68–95–99.7% rule

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$P(|X-\mu|<\sigma) \approx 0.68, \qquad P(|X-\mu|<2\sigma) \approx 0.95, \qquad P(|X-\mu|<3\sigma) \approx 0.997.$$

That is, the chance of falling within 1, 2, 3 SDs of the mean.

Example 5.26 (Using the 68–95–99.7 rule)

Let $X \sim \mathcal{N}(-1, 4)$. Approximate $P(|X|<3)$.

Solution. Here $\mu = -1$ and $\sigma = 2$. $|X|<3$ is the same as $-3<X<3$. Standardize with $Z = \frac{X-(-1)}{2}$:

$$P(-3<X<3) = P\left(\frac{-3-(-1)}{2} < \frac{X-(-1)}{2} < \frac{3-(-1)}{2}\right) \qquad \text{standardize}$$

$$= P(-1<Z<2).$$

The rule says $P(-1<Z<1) \approx 0.68$ and $P(-2<Z<2) \approx 0.95$, so the two bands between 1 and 2 SDs hold $0.95-0.68 = 0.27$ together, and by symmetry $P(1<Z<2) \approx 0.27/2$. So

$$P(-1<Z<2) = P(-1<Z<1) + P(1<Z<2) \approx 0.68 + 0.135 = 0.815.$$

The exact value is $\Phi(2)-\Phi(-1) \approx 0.8186$, so the rule does well.

Example 5.27 (Folded Normal)

Let $Y = |Z|$ with $Z \sim \mathcal{N}(0,1)$. Find (a) $\mathrm{E}(Y)$, (b) $\mathrm{Var}(Y)$, (c) the CDF and PDF of Y.

Solution. (a) By LOTUS, and since $|z|\varphi(z)$ is even,

$$\mathrm{E}(Y) = \int_{-\infty}^{\infty} |z| \, \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z\, e^{-z^2/2}\, dz \qquad \text{even function}$$

$$= \frac{2}{\sqrt{2\pi}} \Big[-e^{-z^2/2}\Big]_0^{\infty} = \frac{2}{\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}} \approx 0.80.$$

(b) $\mathrm{E}(Y^2) = \mathrm{E}(Z^2) = \mathrm{Var}(Z) = 1$, so

$$\mathrm{Var}(Y) = \mathrm{E}(Y^2) - (\mathrm{E}\,Y)^2 = 1 - \frac{2}{\pi} \approx 0.36.$$

(c) For $y \geq 0$,

$$F_Y(y) = P(|Z| \leq y) = P(-y \leq Z \leq y) = \Phi(y) - \Phi(-y) = 2\Phi(y) - 1,$$

using $\Phi(-y) = 1 - \Phi(y)$, and $F_Y(y) = 0$ for $y < 0$. Differentiating,

$$f_Y(y) = 2\varphi(y) = \sqrt{\frac{2}{\pi}}\, e^{-y^2/2}, \qquad y > 0.$$

5.5 Exponential

The Exponential is the continuous counterpart of the Geometric. A Geometric r.v. counts the number of failures before the first success in a sequence of Bernoulli trials. With the Exponential, we're waiting for a success in *continuous time*, where successes arrive at a rate of λ per unit of time. The average number of successes in a time interval of length t is λt, though the actual number varies randomly. An Exponential r.v. represents the waiting time until the first arrival.

Definition 5.28 (Exponential Distribution)

A continuous r.v. X has the **Exponential distribution** with parameter (rate) $\lambda > 0$, written $X \sim \text{Expo}(\lambda)$, if its PDF is

$$f(x) = \lambda e^{-\lambda x}, \qquad x > 0.$$

This is valid since $\int_0^\infty \lambda e^{-\lambda x}\, dx = 1$. The CDF is

$$F(x) = \int_0^x \lambda e^{-\lambda t}\, dt = 1 - e^{-\lambda x}, \qquad x > 0.$$

Proposition 5.29 (Scaling to Expo(1))

If $X \sim \text{Expo}(\lambda)$, then $Y = \lambda X \sim \text{Expo}(1)$.

Proof. For $y > 0$,

$$P(Y \leq y) = P\Big(X \leq \frac{y}{\lambda}\Big) = 1 - e^{-\lambda \cdot y/\lambda} = 1 - e^{-y},$$

which is the Expo(1) CDF. □

Theorem 5.30 (Exponential Mean and Variance)

If $X \sim \text{Expo}(\lambda)$, then

$$\text{E}(X) = \frac{1}{\lambda}, \qquad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Proof. First do $Y \sim \text{Expo}(1)$, using integration by parts each time.

First moment. With $u = y$, $dv = e^{-y}dy$ (so $du = dy$, $v = -e^{-y}$):

$$\mathrm{E}(Y) = \int_0^\infty y\, e^{-y}\, dy = \Big[-y\, e^{-y}\Big]_0^\infty + \int_0^\infty e^{-y}\, dy = 0 + 1 = 1.$$

Second moment. With $u = y^2$, $dv = e^{-y}dy$:

$$\mathrm{E}(Y^2) = \int_0^\infty y^2 e^{-y}\, dy = \Big[-y^2 e^{-y}\Big]_0^\infty + 2\int_0^\infty y\, e^{-y}\, dy = 0 + 2\ \mathrm{E}(Y) = 2,$$

so $\mathrm{Var}(Y) = 2 - 1^2 = 1$.

General λ. By Proposition 5.29, $X = Y/\lambda$ with $Y \sim \mathrm{Expo}(1)$, so

$$\mathrm{E}(X) = \frac{\mathrm{E}(Y)}{\lambda} = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{\mathrm{Var}(Y)}{\lambda^2} = \frac{1}{\lambda^2}.$$

□

Pitfall. λ is a *rate*, not a mean. If buses come on average every 10 minutes, then $\lambda = 1/10$ per minute, not 10. (Some books parametrize the Exponential by its mean instead, so always check.)

Definition 5.31 (Memoryless Property)

A continuous distribution on $(0, \infty)$ is **memoryless** if, for X with that distribution,

$$P(X \geq s + t \mid X \geq s) = P(X \geq t) \quad \text{for all } s, t \geq 0.$$

Think of X as a waiting time (e.g. for a phone call): if you've already waited s minutes, the chance of waiting at least t *more* minutes is the same as if you'd just started.

Theorem 5.32 (The Exponential is Memoryless)

If $X \sim \mathrm{Expo}(\lambda)$, then X has the memoryless property.

Proof. Here $P(X \geq s) = 1 - P(X \leq s) = e^{-\lambda s}$. Then

$$P(X \geq s + t \mid X \geq s) = \frac{P(X \geq s + t,\ X \geq s)}{P(X \geq s)} \qquad \text{def. of cond. prob.}$$

$$= \frac{P(X \geq s + t)}{P(X \geq s)} \qquad X \geq s \text{ is redundant}$$

$$= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X \geq t).$$

□

Example 5.33 (Waiting for a bus)

Bus arrivals at a stop follow a rate of one every 10 minutes on average, so your wait is $X \sim \text{Expo}(1/10)$ in minutes. You've already waited 10 minutes. What's the probability you wait more than 15 minutes in total?

Solution. By memorylessness, having waited 10 minutes changes nothing:

$$P(X > 15 \mid X > 10) = P(X > 5) = e^{-5/10} = e^{-0.5} \approx 0.61.$$

Your expected *remaining* wait is still $1/\lambda = 10$ minutes, no matter how long you've been standing there.

Theorem 5.34 (Memoryless Implies Exponential)

If X is a positive continuous r.v. with the memoryless property, then $X \sim \text{Expo}(\lambda)$ for some $\lambda > 0$.

Proof. Let F be the CDF of X and $G(x) = P(X > x) = 1 - F(x)$. The memoryless property says

$$G(s + t) = G(s)\, G(t) \quad \text{for all } s, t \geq 0.$$

We solve for G.

- Taking $s = t$ gives $G(2t) = G(t)^2$, then $G(3t) = G(t)^3$, and in general $G(kt) = G(t)^k$ for positive integers k.
- Replacing t by t/k: $G(t) = G(t/k)^k$, so $G(t/k) = G(t)^{1/k}$.
- Combining the two: $G\big(\frac{m}{n}t\big) = G(t)^{m/n}$ for all positive integers m, n, so $G(xt) = G(t)^x$ for every positive rational x.
- Any positive real number is a limit of positive rationals, and G is continuous, so $G(xt) = G(t)^x$ for all real $x > 0$.

Taking $t = 1$: $G(x) = G(1)^x = e^{-\lambda x}$, where $\lambda = -\ln G(1)$. Since X is positive and continuous, $0 < G(1) < 1$, so $\lambda > 0$. Then $F(x) = 1 - e^{-\lambda x}$, the $\text{Expo}(\lambda)$ CDF. □

5.6 Poisson Processes

The Poisson and the Exponential both use λ for a reason: they're two sides of the same story. Think of emails landing in your inbox at random times. You can ask two kinds of questions:

- **Counting:** how many emails arrive in the next hour? That's a whole number, so a *discrete* r.v. (Poisson).

- **Waiting:** how long until the next email? That's a positive real number, so a *continuous* r.v. (Exponential).

A Poisson process is the model where both answers work out nicely at the same time.

Definition 5.35 (Poisson Process)

A process of arrivals in continuous time is a **Poisson process with rate** λ if:

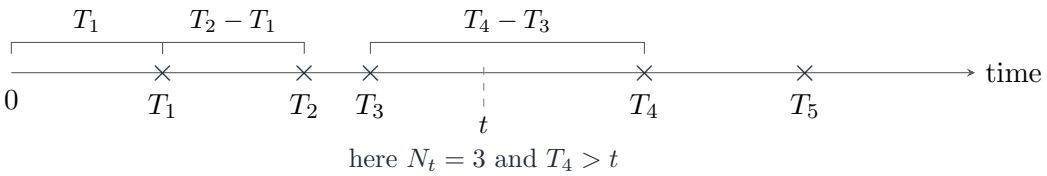
1. the number of arrivals in any interval of length t is $\text{Pois}(\lambda t)$;
2. the numbers of arrivals in disjoint (non-overlapping) intervals are independent.

Intuition. λ is the *rate*: the average number of arrivals per unit of time. In an interval of length t you expect λt arrivals, but the actual count is random. Condition 2 says the process has no memory between time windows: how many emails came between 9 and 10 tells you nothing about how many come between 10 and 11.

Notation

For a Poisson process on $(0, \infty)$:

- N_t = number of arrivals in $(0, t]$, so $N_t \sim \text{Pois}(\lambda t)$;
- T_n = time of the nth arrival;
- $T_1,\ T_2 - T_1,\ T_3 - T_2, \ldots$ = the **interarrival times** (gaps between arrivals).



A Poisson process: each $\times$ is an arrival. The gaps between them are the interarrival times.

Count-time duality

$$T_1 > t \iff N_t = 0, \qquad T_n > t \iff N_t < n.$$

In words: "the nth arrival hasn't happened by time t" is the *same event* as "fewer than n arrivals by time t". Since they're the same event, they have the same probability. This is the trick that lets us go between counts (discrete) and times (continuous).

Theorem 5.36 (Interarrival Times are i.i.d. Exponential)

In a Poisson process with rate λ, the time of the first arrival is $T_1 \sim \text{Expo}(\lambda)$. More generally, the interarrival times $T_1, T_2 - T_1, T_3 - T_2, \ldots$ are i.i.d. Expo(λ).

Proof. For $t > 0$,

$$P(T_1 > t) = P(N_t = 0) \qquad \text{count-time duality}$$

$$= \frac{e^{-\lambda t}(\lambda t)^0}{0!} = e^{-\lambda t}, \qquad N_t \sim \text{Pois}(\lambda t)$$

so $P(T_1 \leq t) = 1 - e^{-\lambda t}$, which is the Expo($\lambda$) CDF.

For the gaps: once the first arrival happens, what happens next only depends on arrivals in later, disjoint intervals, which are independent of everything before (condition 2). So the process "starts fresh" at T_1, and by the same argument $T_2 - T_1 \sim \text{Expo}(\lambda)$, independent of T_1. Repeating this at each arrival, all the gaps are i.i.d. Expo(λ). □

Remark. So the Poisson process ties the two distributions together: **counts** in an interval are Poisson, **gaps** between arrivals are Exponential, and λ is the same rate in both. Starting fresh at each arrival is the memoryless property in action.

Pitfall. Only the *gaps* are Exponential. The time of the nth arrival, T_n, is a *sum* of n i.i.d. Expo(λ) gaps, and for $n \geq 2$ it is not Exponential. (It has a Gamma distribution, which comes in Chapter 8.)

Example 5.37 (Emails as a Poisson process)

Emails arrive as a Poisson process with rate $\lambda = 6$ per hour. Find (a) the probability of no emails in the next 20 minutes, (b) the expected time until the 3rd email, and (c) $P(T_2 > t)$, the probability the 2nd email hasn't arrived by time t.

Solution. Work in hours, so 20 minutes is $t = 1/3$ and $\lambda t = 2$.

(a) Two ways, same answer. With counts: $N_{1/3} \sim \text{Pois}(2)$, so $P(N_{1/3} = 0) = e^{-2}$. With times: $T_1 \sim \text{Expo}(6)$, so $P(T_1 > 1/3) = e^{-6/3} = e^{-2} \approx 0.135$.

(b) T_3 is the sum of three Expo(6) gaps, so by linearity

$$\text{E}(T_3) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2} \text{ hour} = 30 \text{ minutes}.$$

(c) Use count-time duality with $n = 2$:

$$P(T_2 > t) = P(N_t < 2) = P(N_t = 0) + P(N_t = 1) \qquad \text{count-time duality}$$

$$= e^{-6t} + 6t\, e^{-6t} = e^{-6t}(1 + 6t). \qquad N_t \sim \text{Pois}(6t)$$

This isn't of the form e^{-ct}, which confirms T_2 is not Exponential.

Proposition 5.38 (Minimum of Independent Exponentials)

Let $X_1, \ldots, X_n$ be independent with $X_j \sim \text{Expo}(\lambda_j)$, and let $L = \min(X_1, \ldots, X_n)$. Then

$$L \sim \text{Expo}(\lambda_1 + \cdots + \lambda_n).$$

Proof. Work with the survival function $P(L > t)$, which is $1-$ the CDF:

$$P(L > t) = P(X_1 > t, \ldots, X_n > t) \qquad \min > t \text{ iff all } > t$$

$$= P(X_1 > t) \cdots P(X_n > t) \qquad \text{independence}$$

$$= e^{-\lambda_1 t} \cdots e^{-\lambda_n t} = e^{-(\lambda_1 + \cdots + \lambda_n)t},$$

which is the survival function of $\text{Expo}(\lambda_1 + \cdots + \lambda_n)$. □

Intuition. Picture n independent Poisson processes running at once: red cars pass at rate λ_1, blue cars at rate λ_2, and so on. X_j is the wait for the first car of colour j, and L is the wait for the first car of *any* colour. All the cars together arrive at the combined rate $\lambda_1 + \cdots + \lambda_n$, so of course L is Exponential with that rate.

Pitfall. The *minimum* of independent Exponentials is Exponential, but the *maximum* is not. The trick for the max is to break it into a sum of minimums, as in the next example.

Example 5.39 (Maximum of three Exponentials)

Three independent servers each run for an $\text{Expo}(\lambda)$ amount of time with mean 12 months, then fail. Let T be the time until *all three* have failed. Find $\text{E}(T)$.

Solution. Here $\lambda = 1/12$ per month and $T = \max(X_1, X_2, X_3)$. T is not Exponential: its CDF is $P(T \le t) = (1 - e^{-\lambda t})^3$. Instead of integrating, split T into three stages:

$$T = \underbrace{T_1}_{\text{1st failure}} + \underbrace{T_2}_{\text{1st to 2nd}} + \underbrace{T_3}_{\text{2nd to 3rd}}.$$

- T_1 is the min of three $\text{Expo}(\lambda)$'s, so $T_1 \sim \text{Expo}(3\lambda)$ by Proposition 5.38.

- At the first failure, the other two servers start fresh (memoryless), so T_2 is the min of two Expo(λ)'s: $T_2 \sim \text{Expo}(2\lambda)$.
- Likewise, $T_3 \sim \text{Expo}(\lambda)$.

By linearity,

$$\text{E}(T) = \frac{1}{3\lambda} + \frac{1}{2\lambda} + \frac{1}{\lambda} = 4 + 6 + 12 = 22 \text{ months}.$$

Notice it's not 3×12: once one server fails, the remaining ones don't have to "start over".

Example 5.40 (A machine that breaks and gets fixed)

A machine works for an Expo(λ) time, breaks, takes an Expo(λ) time to fix, works again for an Expo(λ) time, and so on, with all these times i.i.d. (in days).

(a) A *transition* is a switch between working and broken. Find the distribution of the number of transitions in $(0, t)$.

(b) The machine is redesigned with 5 components, each failing after an independent Expo(λ) time. It keeps working as long as at most one component has failed. Starting with all 5 working, find the expected time until the machine breaks.

Solution. (a) The times between transitions are i.i.d. Expo(λ), so the transitions form a Poisson process with rate λ. The number in $(0, t)$ is Pois(λt).

(b) The machine breaks at the *second* component failure. The first failure is the min of 5 Exponentials, so it takes Expo(5λ) time. After that, by memorylessness, the other 4 start fresh, so the next failure takes Expo(4λ) more. By linearity,

$$\text{E}(\text{time until breakdown}) = \frac{1}{5\lambda} + \frac{1}{4\lambda} = \frac{9}{20\lambda}.$$

Poisson process cheat sheet

- **Counts:** arrivals in an interval of length $t \sim \text{Pois}(\lambda t)$; disjoint intervals are independent.
- **Gaps:** interarrival times are i.i.d. Expo(λ), each with mean $1/\lambda$.
- **Count-time duality:** $T_n > t \iff N_t < n$. Use it to turn a question about times into one about counts.
- **Min of Expos:** independent Expo(λ_j)'s have a minimum that is Expo$\left(\sum_j \lambda_j\right)$.
- **Max of Expos:** not Exponential. Break it into stages with min + memorylessness, then use linearity.